

INTERN GUIDE

Crypto Exchange Knowledge Handbook

From fundamentals to practice,
understanding digital asset trading platforms

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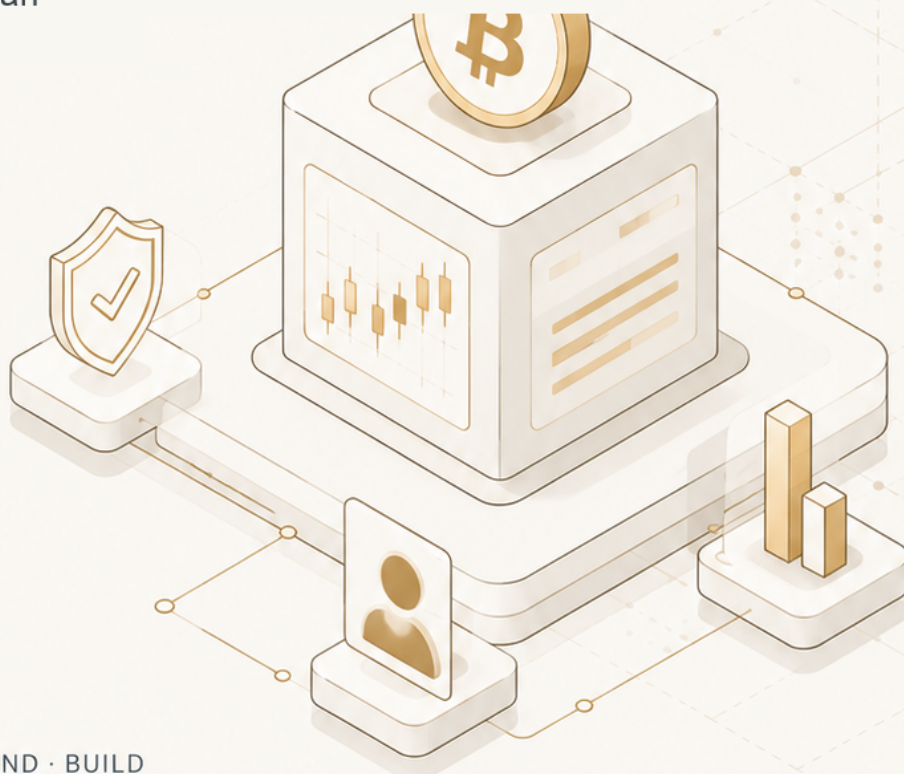


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Preface

This handbook began as an introductory guide I wrote for interns on my team in late 2018. Its original purpose was simple: to help new colleagues build the industry context, platform knowledge, and working vocabulary they needed to collaborate effectively across teams.

In early 2026, at the request of a Hong Kong investment bank, I undertook a systematic revision of the manuscript. Two industry friends reviewed the draft; they currently serve as a vice president for trading products and a chief security officer at two leading crypto exchanges. I am grateful for their comments and support.

This handbook is intended as entry-level training material. It is designed for rapid onboarding and cross-functional collaboration, not as a substitute for professional training in trading technology, cybersecurity, or compliance. After reading it, readers should be able to follow most business discussions and know which colleagues to consult when a topic requires deeper expertise.

Given the speed of the revision and the limits of my own experience, omissions are inevitable. I welcome corrections and criticism.

If you have any questions or comments, please contact the author at bond.tian@protonmail.com.

Bond TIAN

About the Author: Bond Tian (田邦德) is a former senior executive at Huobi and OKX, where he worked across platform operations, marketing, government relations, and related business functions. Drawing on ten years of hands-on experience in the crypto exchange industry, he wrote this comprehensive handbook for university students and early-career professionals interested in understanding how crypto trading platforms actually operate. The handbook is designed as a rare practitioner-written learning guide, built from firsthand exchange experience rather than secondhand observation.

**The incident cases discussed in this handbook are compiled from public sources. They are used only to help readers understand terminology, technical logic, and operational risk, and should not be read as final determinations of the full causes of any incident.*

How to Read This Handbook: Scope and Audience

This handbook is an entry-level training guide based on practical experience inside crypto exchanges. It explains how crypto trading platforms operate end to end: how they match orders, custody assets, generate revenue, manage compliance and security risks, and run growth, token listing, customer support, and on-chain operations.

Readers without a background in finance may find some sections challenging. If you are based in mainland China, taking the securities practitioner qualification exam first can provide useful grounding.

Suitable for readers who:

- College students who want to enter the crypto industry
- New entry-level practitioners
- Users who are already trading on the exchange and want to understand how the platform operates

Difficulty levels: The handbook is organized into three levels of difficulty. Readers can choose a reading path based on their prior knowledge:

Level	Chapter	Suitable for whom	Core Takeaways
Basics	Chapter 1, 5, 6	Beginner readers, ordinary users	Understand what an exchange is, how it makes money, and what its core products are
Core Operations	Chapter 2, 3, 7, 10, 11, 15	Junior practitioners, operations/customer support/risk management roles	Understand matching, wallet security, liquidation risk management, on-chain deposits and withdrawals, token listing and customer support and complaint handling
Operations and Compliance	Chapter 4, 8, 12, 13, 14	Manager, compliance/finance/institutional business positions	Understand licenses, financial models, stablecoins, data operations and market maker cooperation

Level	Chapter	Suitable for whom	Core Takeaways
Industry Extension	Chapters 16 - 19	Readers who want to build an industry knowledge system	Understand prediction markets, RWA, U-cards, crypto payments, public chains, mining, mines and mining pools

Timeliness note: The crypto industry changes quickly. The core of this handbook is based on the original draft from late 2018, while sections involving regulation, licensing, exchange market structure, and major cases were reviewed against public information available in early 2026. Before using this material for formal training, investment, partnership, or compliance decisions, verify the latest regulator notices, company disclosures, and professional advice.

Reading suggestion: Read the chapters in order, as later sections repeatedly build on earlier concepts. The opening note in each chapter highlights the key ideas, and the chapter summary helps review the main takeaways.

Chapter 1 Overview and Business Model of Crypto Exchanges

Chapter Overview

This chapter answers three questions: what a crypto exchange is, what types of exchanges exist, and how exchanges make money. By the end, readers should understand the basic role of an exchange and the main differences between CEX, DEX, hybrid exchanges, and brokerage platforms.

1.1 What Is an Exchange? Start with the Word Itself

In short, a crypto exchange is a platform that provides order-matching services. A centralized exchange (CEX) also custodies user assets, while a decentralized exchange (DEX) usually leaves assets in users' own wallets. Some hybrid DEX models still include centralized components, which are discussed later. In traditional markets, an exchange centralizes quotes, matches buyers and sellers, maintains trading order, and charges fees. Crypto exchanges play a similar role, but the traded assets are Bitcoin, Ethereum, stablecoins, and other digital assets rather than stocks, commodities, or foreign exchange.

The key difference is custody. On a CEX, assets deposited by users usually enter a wallet system controlled by the platform, and users see account balances recorded in the platform ledger. On a DEX, trades are executed through on-chain mechanisms. Some DEXs use automated market makers (AMMs) and liquidity pools for pricing and execution, such as Uniswap; others use on-chain order books, such as dYdX v4 after its migration to a Cosmos appchain. In most DEX models, assets remain in the user's own wallet.

Practitioner note: Withdrawing coins from an exchange to your own wallet is essentially a move from platform custody to self-custody. This reduces platform custody risk, but it also means the user becomes responsible for private key management, phishing prevention, address verification, and transaction risk.

1.2 Four Common Exchange Models

Digital asset exchanges can be grouped into four broad operating models:

Type 1: Centralized Exchange (CEX) – “Bank + Stock Exchange”

Representatives: Binance, Coinbase, OKX How it works:

1. Users register an account on the exchange and complete identity verification (KYC)
2. Users deposit fiat currencies or cryptocurrencies into exchange accounts
3. Users submit buy/sell orders and the exchange system matches the counterparty

4. After the transaction is completed, the assets are stored in the wallet of the exchange
5. After the user submits a withdrawal application, the exchange will transfer the assets to the user's designated wallet.

Advantages: Fast speed (the peak processing capacity of the matching engine of the leading exchange can reach more than one million transactions per second), good experience (there is an app and customer support), and it supports fiat deposits. Disadvantages: Users need to bear the risk of platform custody. If the exchange is hacked, shuts down, is suspected of misappropriating assets, or freezes accounts, user assets may suffer heavy losses. Case: In November 2022, FTX went from a leading trading platform to bankruptcy and reorganization in a very short period of time. Bankruptcy documents and public reports showed that there was a huge gap in its user assets. Chapter 9 will further analyze this case.

Type 2: Decentralized Exchange (DEX) – “Non-Custodial On-Chain Trading”

Representative: Uniswap, Curve Operation method:

1. Users usually do not need to register an account, only a crypto wallet (such as MetaMask)
2. The assets are always in the user's own wallet and do not need to be "charged" to any platform.
3. Transactions are completed through smart contracts (which can be understood as "self-executing electronic contracts")
4. Transaction results are recorded directly on the blockchain and can be viewed by anyone

Advantages: Users have stronger control over assets and do not need to hand over assets to centralized platforms for custody. Disadvantages: slow speed (each transaction needs to be confirmed on the blockchain), high experience threshold (need to understand wallet and private key management), and usually does not directly support fiat purchases. Before choosing a self-custody wallet, you should first master security mechanisms such as mnemonic phrases, hardware wallets, authorization management, and phishing prevention. Practitioner note: The core advantage of DEX is "no custody required" – users do not need to hand over assets to the platform for safekeeping, so the risk of the platform stopping operations or misappropriating assets is low. But the price is more complex and slower operation.

Type 3: Hybrid Exchange – “Layered Matching and Settlement”

Representatives: Vertex, dYdX v3 (v3 is a hybrid mode, v4 has been migrated to fully decentralized) How it works:

- Match orders at the speed of a centralized exchange (completed on the server, very fast)
- Settlement is handled by a decentralized exchange and is completed on-chain in a verifiable way. This model tries to balance matching efficiency, on-chain verifiability, and user control of assets, but the actual degree of decentralization depends on the design of the order book, clearing process, custody model, oracle system, and governance permissions.

Type 4: Brokerage Platform – “Aggregated Trading and Introducing Brokerage”

Representatives: Robinhood, eToro, the former Huobi Cloud, and platforms such as ChainUp. How it works:

- The platform does not necessarily build a complete exchange system by itself, but helps users execute transactions in other exchanges, liquidity providers or white label systems. Some such platforms clearly state that they are brokers, and some platforms are packaged as self-operated brand exchanges on the front end, but the underlying trading system, wallet system or liquidity may be provided by white label technology providers (such as ChainUp, B2Broker). The brand side is responsible for promotion and user operations on the front end, earning trading fees or price differences, which is essentially closer to the traditional brokerage or introducing brokerage model. Features: Some emerging brokerage platforms support one-stop investment portals for stocks, funds, crypto assets, etc., which are suitable for users who want to manage multiple types of assets in the same app.

1.3 How Do Exchanges Make Money? Eight Major Revenue Sources

Exchange revenue does not come only from trading fees. It can also include withdrawal fees, staking products, futures, Launchpad services, data services, institutional services, and other value-added products. The main revenue sources are broken down below by business line:

Revenue Source 1: Trading Fees

Every time a user buys or sells on an exchange, the platform charges a trading fee, often around 0.02% to 0.2% of the trade amount. On trading-focused platforms, these fees are usually one of the most important revenue sources, although the exact share varies by platform type, market cycle, and product mix. For example, if a user buys \$10,000 of Bitcoin at a 0.1% fee rate, the fee is \$10. A single trade may look small, but on high-volume platforms, trading fees add up quickly. Maker and taker fees: these terms often appear on exchange fee pages. A maker places a limit order that does not execute immediately and instead rests in the order book, adding liquidity. A taker trades against an existing order and removes liquidity from the order book.

Practitioner note: Exchanges usually give Makers lower fees, and in some cases even rebates, because Makers improve market depth and user trading experience.

Revenue Source 2: Listing Fees

If a new project wants to be listed on an exchange, it will usually incur a series of costs related to listing, including technical access, legal opinions, contract audits, market making deposits, marketing resources, and ongoing operational cooperation. Historically, some project teams have disclosed higher costs for listing, but compliance platforms should generally not regard simple payment as the only or decisive condition for listing. Being listed on a large exchange means higher exposure, more liquidity sources and stronger market

endorsement, but it will also bring higher information disclosure and continuous monitoring requirements.

Revenue Source 3: Withdrawal Fees

On-chain transfers require network fees paid to miners or validators. When users withdraw crypto from an exchange to their own wallets, the exchange usually charges a withdrawal fee on top of the network cost. Some exchanges use floating fees based on the actual gas fee plus a margin; others use fixed fees. For major assets such as BTC and ETH, withdrawal fees are often a low-margin service. For long-tail assets, however, withdrawal pricing may create a spread between the platform fee and the actual network cost.

Revenue Source 4: Margin and Lending Interest

Leveraged trading uses margin to magnify trading exposure. The user posts margin of \$1,000, borrows or takes the equivalent exposure of \$9,000, and trades at a total size of \$10,000. When the price rises by 10%, the initial capital may double; when the price falls by 10%, the initial capital may be lost entirely. When the platform provides funds or leveraged exposure, interest or related fees will be charged. The specific rates vary depending on the platform and market conditions.

Revenue Source 5: Staking Revenue Share

Some cryptocurrencies, such as ETH, use a “proof-of-stake” mechanism. Users stake their assets to help maintain the network, and the network issues rewards to validators. The exchange can complete staking, node operation and income distribution on behalf of users, and draw a certain service fee from the income.

Revenue Source 6: Futures Trading Fees and Funding Rates

A perpetual contract is a futures contract with no expiration date that allows users to go long or short using leverage. In order to keep the contract price consistent with the spot price, the perpetual contract sets a "funding rate"

Mechanism: When the market is bullish, the long side usually pays the short side and vice versa. Funding rates mainly flow between long and short parties, while exchanges make profits through trading fees such as opening and closing positions. Since leverage amplifies notional trading volume, futures fee income is usually higher than spot trading.

Revenue Source 7: IEO and Launchpad Service Fees

IEO (Initial Exchange Offering) is an exchange selling tokens for new projects, similar to "new stock subscription." Exchanges charge sales commissions or token shares.

Revenue Source 8: API and Data Services

API (application Programming Interface) is a program interface that allows programs to "talk". The quantitative trading team obtains real-time market conditions, places orders, etc. through API. The exchange sells value-added services such as high-frequency market data and in-depth data to institutional clients.

Case: Breakdown of Coinbase's 2023 revenue structure. The following takes the data disclosed in Coinbase's 2023 shareholder letter as an example to illustrate how the revenue structure of compliant exchanges has moved from a single trading fee to diversification. Coinbase's total revenue for 2023 will be approximately US\$3.1 billion, including approximately US\$1.5 billion in trading revenue and approximately US\$1.4 billion in subscription and service revenue. The two types of income are close to the same level.

Trading revenue mainly comes from fees paid by retail and institutional users when they buy and sell crypto assets. Retail fee rates are usually higher, so even when retail users do not account for the largest share of total volume, they may still contribute more revenue per unit of volume.

Subscription and service income includes blockchain rewards (Staking), stablecoin-related income, custody fees, interest income and other service income. This type of revenue is characterized by a low correlation with trading volume, which helps smooth revenue fluctuations in bear markets.

Key Takeaways

- Retail investors generally contribute higher unit fee income
- Subscription and service revenue can reduce the platform's reliance on short-term trading volume
- Stablecoin, custody, staking and interest-related businesses are becoming important sources of income for compliant exchanges

Long-Tail Asset Strategies for Small and Mid-Sized Exchanges

Platforms such as MEXC (Matcha) and Gate often form differentiated positioning outside of large platforms such as Binance and OKX by supporting long-tail assets faster.

Strategy: According to the snapshot of the data website at a certain point in time, small and medium-sized exchanges tend to list more long-tail assets, and the number may be significantly higher than that of the leading compliance platforms. The new coins, small coins, and meme coins that users are looking for may be supported by small and medium-sized exchanges earlier.

Cost: The depth of liquidity is far less than that of the head platform. When users sell larger amounts of long-tail assets, they may have to endure slippage of 1-2% or even higher; the quality of some listed coins is also uneven, and the risk of being caught is high.

Lesson: Small and medium-sized exchanges usually use asset coverage speed and low rates to attract traffic. But in the long term, token listing quality, real liquidity and risk management are still core competitiveness.

In addition, exchanges usually set up self-operated funds or affiliated funds to participate in primary market investments. This is a means of profit for the exchange. At the same time, it is also a strategic work to obtain cutting-edge information and project resources in the primary market. This is also an important means to maintain the influence of the trading platform in the industry.

It is particularly important to point out that in the traditional securities market, trading venues, self-operated securities firms, investment banking businesses and related investments are usually subject to strict information isolation, conflicts of interest, related transactions and capital rules. "Platform influence" and "investment income" cannot be simply confused.

There is still no single global regulatory framework for the crypto industry, and many markets have not yet legislated clearly. As regulation tightens, however, conflicts of interest between exchanges and affiliated investment or trading arms are receiving more scrutiny, including from the U.S. SEC and under the EU MiCA framework. The collapse of FTX exposed the systemic risks of related-party structures: Alameda Research, as an affiliated market maker and trading firm, became a textbook example of blurred boundaries. These restrictions are likely to become stricter as regulatory frameworks mature.

1.4 What Does an Exchange Look Like? A Full Organizational Map

The previous section talked about “what” an exchange does, and this section talks about “how it is organized”. Understanding the organizational structure can help you understand how departments work together and which teams you might work on in the future.

Exchange Organization at a Glance

The above is a panoramic view of the exchange’s organizational structure. The first layer coordinates the overall situation; the second layer has CTO, COO, CCO, CFO, and CISO in charge of core functions; the third layer is responsible for execution; the fourth layer undertakes specific professional tasks; and the fifth layer forms business outputs for users and institutional customers.

The Five-Layer Operating Architecture

First level: CEO (Chief Executive Officer)

The exchange's top decision-maker is responsible for:

- Strategic direction: deciding which markets to enter and which products to launch
- External relations: dealing with regulators, investors, and partners
- Overall management: Coordinate resources of various departments to ensure company operation

Second level: C-Level executive team

A mature exchange usually has 5 core executive positions:

Position	Main responsibilities	operational significance
CTO (Chief Technology Officer)	Responsible for the matching engine, wallet system, front-end/mobile terminal, back-end services and system operation and maintenance	Technical stability directly determines trading experience and asset security
COO (Chief Operating Officer)	Responsible for marketing, user operations, customer support, community and event growth	Keep exchange users, liquidity and product usage running
CCO (Chief Compliance Officer)	Responsible for compliance, legal affairs, KYC/AML, license application and regulatory docking	As supervision becomes more stringent, compliance capabilities have become a long-term operating threshold.
CFO (Chief Financial Officer)	Responsible for financial accounting, fund management, on-chain fund scheduling, insurance funds and tax planning	Client assets, company assets and risk reserves must be clearly defined
CISO (Chief Security Officer)	Responsible for information security, asset security, penetration testing, emergency response, smart contract audit and on-chain monitoring	Major security incidents may directly affect the platform's ability to continue operating

The third level: core departments

There are several departments under each C-Level executive, covering several core functional departments (see the organizational structure table for details).

Level 4: Segmentation Teams

The departments are further subdivided into specific execution teams. For example:

- Wallet system group → hot wallet operation and maintenance, cold wallet management, multi-chain adaptation
- Compliance review → KYC/AML review, license application
- Security Attack and Defense → Penetration Testing, Emergency Response

Level 5: Business units

This is the line of business that directly generates revenue:

- Spot trading, futures trading, Earn/staking products, fiat channel, Web3 wallet, institutional services

How Many People Work at an Exchange?

Exchanges of different sizes have huge differences in personnel. The following data is the author's guess based on industry experience:

Ultra-large (Binance, OKX): 5,000–8,000 people (as of May 2025) – Technical team accounts for 40–50% – Operations/Customer Service accounts for 20–30% – Compliance/Legal Affairs accounts for 10–15% – Others (administration, HR, etc.) account for 10–15%

Medium (Bitget, KuCoin): 500–2,000 people – Technical team accounts for a higher proportion (maybe up to 60%) – Operations team is relatively lean

Small/start-up: 50–200 people – Multiple jobs per person is common – Compliance and security may be outsourced

Tip: Technical positions (back-end development, security engineers) are the most in-demand entry-level positions at exchanges. Operations and compliance positions require more industry knowledge and language skills because they need to serve global users.

In addition, it needs to be pointed out that some exchanges have separation of front and back offices and blurred organizational structures. Executives who frequently attend public events may not have actual operating responsibilities; common titles include CEO, CMO, business VP, strategy VP, or nominal investment researcher. This arrangement will seriously affect industry recruitment judgment, business cooperation and responsibility identification. Students should pay special attention to the actual decision-making chain when conducting due diligence and business cooperation.

Chapter Summary

- Crypto exchanges are a complex of “digital asset banks + brokers + payment networks”. Each of the four types has applicable scenarios.
- The core profit model of the exchange is essentially the realization of liquidity, asset precipitation and risk pricing

- Compliance and security roles are transforming from cost centers to sources of competitiveness

Chapter 2 Core Trading System Architecture

Chapter Overview

This chapter explains the basic structure of an exchange backend: how orders are matched, how assets are custodied, and how the system remains stable under high concurrency.

2.1 Exchange Technical Architecture: The Four-Layer Stack

A mature exchange system can be divided into four layers. To facilitate reading in Word, the following is presented in a table:

Hierarchy	Name	Main modules	Popular understanding
first floor	access layer	App, web page, API, WebSocket	Entrance to the exchange for users and program traders
second floor	business layer	Registration, login, KYC, deposit and withdrawal, order submission, risk pre-check	Handle daily business rules and user actions
third floor	core layer	Matching engine, market service, clearing service, wallet service	The core of trading, bookkeeping, clearing and asset management
fourth floor	infrastructure layer	Database, message queue, cache, monitoring and alarming	The underlying capabilities that support the stable operation of the system

What exactly do each layer do?

The first layer: Access layer – "the interface between you and the exchange" This layer is what you see: the exchange's web page, mobile app, and API interface for programmatic traders.

Web page/app: When you click "Buy" or "Sell" with your mouse, you are interacting with this layer.

API interface: The quantitative trading team writes a program to automatically place orders, which is to "talk" to the exchange through the API

WebSocket: A "continuous connection" technology that allows exchanges to push price changes to you in real time – just like WeChat messages popping up in real time without you having to refresh manually

The second layer: business layer – "handling daily affairs"

User management: registration, login, password retrieval KYC authentication: upload ID card, face recognition – confirm that you are you. Some regulated and compliant exchanges will also require the same materials as banks, such as real address, utility bills.

Asset management: How much did you deposit in and how much did you withdraw? Risk engine: Check before you place an order – is there anything abnormal in this operation? Are there any violations?

The third layer: core layer – "core transaction module"

Matching engine: Match your buy orders with other people's sell orders – this is the core function of the exchange, which will be discussed in detail later.

Market-data services: generate candlestick charts, depth charts, 24-hour price changes, and related market data. Clearing services: after a trade is completed, assets are transferred from the seller to the buyer. On a centralized exchange, this is usually an internal ledger transfer rather than an on-chain transfer.

Wallet service: Manage the exchange's on-chain wallet—hot wallet, warm wallet, cold wallet

The fourth layer: infrastructure layer – "invisible but inseparable"

Database: stores all user information and transaction records (just like a bank's ledger)

Message queue: allows efficient communication between different systems (like a conveyor belt in a factory)

Cache: keep frequently used data "at hand", without having to check the database every time (just like you save frequently used phone numbers in your mobile phone address book, instead of flipping through the phone book every time) Monitoring and alarming: automatically alarm when there is a problem with the system (like a smoke alarm)

Case: Binance matching engine failure in February 2021 On February 19, 2021, Binance experienced a serious failure and the entire site was down for 4 hours. what happened?

- At that time, the market was highly volatile, and a large number of users submitted orders and liquidation requests at the same time.
- While the matching engine is processing these requests, memory usage rises sharply.
- The operating system found that there was not enough memory and directly "killed" the matching engine process (equivalent to a system-level forced interrupt)
- Result: All users are unable to trade, deposit, or withdraw money for about 4 hours.

What happened during this period?

- Bitcoin price rises from \$56,000 to \$57,500, but Binance users are completely unable to operate
- Users express dissatisfaction on social media
- CZ (Changpeng Zhao, Binance CEO) tweeted to confirm the problem 15 minutes later, and the system was restored 4 hours later

- CZ publicly apologized and announced compensation for affected users. What did Binance do afterwards?
- Restructured the architecture of the matching engine
- Introduced memory usage warning – expand the memory in advance when it is almost full, instead of waiting until it "explodes"
- An automatic expansion mechanism has been added – automatically adding servers when traffic surges. Lesson: Even the top exchanges may have problems with their technical architecture. The key is not to "never be down" (which is almost impossible), but to:
 1. Speed of discovery: Issue discovered within 15 minutes vs. hours later
 2. Recovery speed: Within 4 hours vs. days
 3. Post-event attitude: proactive apology and compensation vs. shirking responsibility.Therefore, the key to a mature exchange is not to promise to "never go down", but to formulate plans in advance, quickly locate problems, restore core services, and clearly communicate compensation standards.

Direct impact on users: BTC price increased by approximately 2.7% during the 4-hour outage. Users holding short positions were unable to close their positions and stop-loss orders, and some users suffered additional losses as a result. This reminds us: During periods of high volatility, the technical stability of the exchange directly affects user profits and losses.

2.2 Matching Engine

What Does the Matching Engine Do?

The matching engine matches buy and sell orders according to predefined price-time priority rules. Think of a used-car market: a seller asks for 100,000, while a buyer offers 95,000. If they cannot agree on price, no trade happens. If a buyer is willing to pay 100,000, the trade is executed. A matching engine does this continuously and in real time, except that instead of one car, it may process millions of buy and sell requests per second.

Order Book: The Table of Bids and Asks

The core data structure of the matching engine is the order book. It records all open buy and sell orders that have not yet been filled. Here is a simplified example:

direction	price	Quantity	cumulative	illustrate
Ask	\$60,300	0.8 BTC	2.5 BTC	Sell orders are arranged from low to high
Ask	\$60,200	1.2 BTC	1.7 BTC	
Ask	\$60,100	0.5 BTC	0.5 BTC	Current lowest selling price
bid-ask spread	\$100			\$60,100 - \$60,000
Bid	\$60,000	1.0 BTC	1.0 BTC	Current highest bid price
Bid	\$59,900	0.3 BTC	1.3 BTC	
Bid	\$59,800	2.0 BTC	3.3 BTC	Buy orders are arranged from high to low

How do you read this table?

- Bids: The current highest bid is \$60,000 - someone is willing to buy 1 BTC for \$60,000
- Asks: The current lowest asking price is \$60,100 - someone is willing to sell 0.5 BTC for \$60,100
- Spread: $\$60,100 - \$60,000 = \$100$. The smaller the price difference, the more active the market and the better the liquidity.

Practitioner Note: Spreads are a direct measure of market liquidity. The BTC/USDT price difference on leading exchanges (such as Binance) is usually only a few dollars; on small and medium-sized exchanges, it may reach tens or even hundreds of dollars.

Matching Rules: Two Non-Negotiables

Rule 1: Price priority - the higher the bid, the sooner the transaction will be completed.

Suppose there are two buy orders:

- Zhang San bids \$60,000 to buy 1 BTC
 - John bids \$60,100 to buy 1 BTC. If a seller sells for \$60,000, John's order will be filled first - because he bids higher.
- Rule 2: Time priority - same price, first come first served

Assume that both Zhang San and Li Si bid \$60,000 to buy 1 BTC, but Zhang San placed the order first. Then when a seller sells for \$60,000, Zhang San's order will be filled first.

Seven Common Order Types

① Limit Order – "I only accept this price" You specify a price, such as "I am only willing to buy BTC at \$60,000." If the current price is above \$60,000, your order is on the order book and so on; if someone is willing to sell it to you for \$60,000, the trade is done. You post a "want to buy" message on Xianyu – "I'll pay 100 yuan for this figure" and wait for the seller to come to you.

② Market Order – "Execute now at the best available price." A market order does not specify a price. It executes immediately against the best available orders in the book. If you want to buy BTC, the system starts from the lowest ask and continues matching until your requested amount is filled. It is like buying a bottle of water at a store without checking the price because you need it right away. Tip: in illiquid markets, market orders can create significant slippage. You may expect to buy at \$60,000, but if there are too few sell orders, the average execution price may become \$60,500.

③ Stop-Loss Order – "Sell if the price falls to this level." You set a trigger price, such as "sell BTC if it falls to \$55,000." When the market price reaches \$55,000, the system submits a market sell order. The common stock-market practice of automatically selling after a 10% drop is essentially a rule-based way to prevent losses from expanding further.

④ Take profit and stop-loss order (TP/SL) – "Sell if it rises here, sell if it falls here" Set two prices at the same time:

- Take Profit: Sell when it reaches \$70,000 – lock in profit
- Stop-Loss: Sell if it falls to \$55,000 – limit losses. You buy a stock and think to yourself, "I will sell it if it rises to 30%, and I will admit it if it falls to 10%." Stop-loss and stop-loss automate this "psychological expectation".

⑤ Iceberg Order – "Only showing a corner" You want to buy 100 BTC, but if you directly place a buy order of 100 BTC, others will find that "there is a big buyer buying" and the price may be raised. So you use an iceberg order – only 5 BTC is displayed on the order book, and then 5 BTC is displayed after the transaction is completed, and so on until you have bought 100 BTC. A rich man went to an auction and didn't want others to know how rich he was, so he sent 10 agents, each of whom would hold up a sign to buy a small portion.

⑥ Post-Only – "I only place orders, not take orders" Make sure that your order must be executed as a Maker (listed on the order book, etc.) rather than immediately executed as a Taker. Why is this so? Because Maker's fees are lower.

⑦ IOC / FOK – "Immediate or Cancel" IOC (Immediate or Cancel): Transact as much as you can, and cancel the rest immediately. For example, if you want to buy 10 BTC, but the current sell order is

only 7 BTC, then buy 7 BTC first, and the order for the remaining 3 BTC will be automatically canceled. FOK (Fill or Kill): Either all transactions are completed or all transactions are cancelled. The same is to buy 10 BTC. If the sell order is only 7 BTC, the entire order will be canceled directly – partial transactions will not be accepted. applicable scenarios: high-frequency trading and arbitrage strategies, which have strict requirements on the number of transactions.

Quick Review: Seven Order Types

type	a word	Suitable for whom
limit order	I only accept this price	Trader who is not in a hurry
market order	No matter how much it costs, I need it now	eager to close a deal
Stop loss order	Sell after falling to the trigger price	Strong awareness of risk management
Take profit and stop loss	Sell when it rises and sell when it falls	Have clear goals
iceberg list	Only a corner exposed	Large households and institutions
Post-Only	Only placing orders but not taking orders	Want to get low Maker rates?
IOC / FOK	Either take it all or lose it	High frequency arbitrage strategy

2.3 Wallet System: Where Are Your Coins Stored?

Three-Layer Wallet Architecture: Hot, Warm, and Cold Wallets

Exchanges manage billions or even tens of billions of dollars in user assets. These assets are not placed in a "wallet", but are scattered in three-tier wallets: Hot Wallet – 24 hours online, processing user withdrawals at any time – Usually only holds 5%-15% of the total assets (varying with exchange size and strategy) – Functions like a cash reserve at a bank counter – ready to respond to customer withdrawal needs – Multi-signature mechanism (at least 2-3 personal signatures can be transferred) – The highest risk: because it is online, it may be hacked

Warm Wallet – Not always online, but can be activated quickly – Stores 20%-30% of total assets – Functions like a hierarchical safe deposit box in a bank branch – requires multiple authorizations to open – Stricter multi-signature (at least 3-5 personal signatures) – Large transfers require handbook approval

Cold Wallet – Completely offline, physically isolated (private keys are stored in hardware devices that are not connected to the Internet) – Stores 60%-70% of total assets – Functions similar to a bank's head office vault – The highest level of asset protection is achieved

through physical isolation - Hardware security module (HSM) protects private keys - Most secure: because it is not connected to the Internet, hackers cannot attack remotely

Key Security Concepts

Multi-Signature - "One lock with multiple keys" Ordinary encryption wallets only need one private key to transfer funds. Multi-signature wallets require simultaneous signatures from several private keys to execute. For example, "3/5 multi-signature": There are 5 private key holders, and at least 3 of them need to agree (sign) to complete the transfer. In this way, even if 1-2 private keys are leaked, the attacker cannot transfer the assets. HSM (Hardware Security Module) - "a safe within a safe" is a dedicated hardware device that can be commonly understood as a computer without an Internet connection. The private key is generated inside the device and is never exported outside the device. All signature operations are completed inside the device - just like a safe. After you put the key in, you can only use it inside the safe and cannot take it out. Address whitelist - "Only transfer to pre-approved addresses" Exchanges can set up a "whitelist" - withdrawals can only be sent to pre-approved addresses. Even if a hacker breaks into your account, he will not be able to transfer the coins to his own address - because his address is not on the whitelist. Summary of this chapter

- The matching engine is the core of the exchange - the order book is its basic data structure, and price priority and time priority are two iron laws.
- The three-layer wallet system (hot/warm/cold) is the basic security framework for exchange asset management
- The reliability of technical architecture—not feature richness—is the basis for an exchange's long-term competitiveness

Chapter 3 Security Architecture and Risk Management

Chapter Overview

Security incidents in the crypto industry can easily cost hundreds of millions of dollars. This chapter uses real cases to understand the security threats faced by exchanges, defense methods, and the hot topic of "proof of reserves."

3.1 What Security Threats Do Exchanges Face?

Security threats to exchanges can be divided into six major categories:

Threat type	Typical risks	Exchange defense line
User account security	Credential stuffing, phishing, 2FA leaks, device manipulation	Device fingerprint, anti-phishing code, login risk checks, withdrawal cooling-off period
Employee permission security	Stolen employee accounts, unauthorized operations, and internal collusion	Permission classification, minimum permissions, approval flow, audit log
Wallet private key security	Hot wallets are stolen, multi-signatures are invalid, and signature processes are hijacked	Hot and cold separation, HSM/MPC, multi-signature, address whitelist
On-chain protocol security	Smart contract vulnerabilities, cross-chain bridge vulnerabilities, oracle machine anomalies	Audit, limit, suspension of deposits and withdrawals, emergency upgrade mechanism
supply chain security	Third-party services, front-end code, and signature interfaces have been tampered with	Integrity verification, supplier review, and secondary confirmation of signature content
Market and operational security	Runs, abnormal liquidation, reputational crisis, customer support mishandling	Emergency plan, insurance fund, unified methodology, upgrade mechanism

3.2 Case Studies

Case 1: 7,000 BTC Stolen from Binance in 2019 – A Typical Phishing Attack

Background: Binance is the largest exchange in the world, and its security system is one of the best in the industry. But on May 7, 2019, it was breached. How did the attacker do it?

According to Binance's announcement at the time and subsequent public reports, the attacker collected a large number of API keys, 2FA codes and related information through long-term phishing, viruses and other attack methods, and transferred 7,000 BTC in a well-organized

withdrawal. The specific attack chain is not completely public, so it cannot be simply understood that a certain employee clicked the wrong email and caused all losses.

The attack process can be broken down into four links:

1. Long-term credential collection: Attackers collect account, API, and authentication information through phishing websites, malware, and more.
2. Bypassing risk management rules: The attacker waits for the right opportunity to combine multiple accounts and withdrawal actions into a centralized attack.
3. A large withdrawal was initiated: 7,000 BTC (approximately \$40 million at the time) was transferred from the hot wallet and split among multiple addresses.
4. Platform emergency response: Binance suspended withdrawals, initiated a safety review, and used SAFU funds to cover user losses.

Result:

- Loss: 7,000 BTC (approximately 2% of Binance' s total assets)
- The assets in the cold wallet (accounting for 90%+ of the total assets) are safe and sound – the hot and cold separation mechanism works
- Binance fully compensates users with SAFU fund (Security Asset Fund for Users)
- Afterwards, Binance fully upgraded its security system.

Lessons:

1. No matter how powerful the technology is, it cannot prevent the leakage of "people" and credentials – attackers may not necessarily crack the core system directly, but may start with accounts, APIs and operating procedures.
2. Hot and cold separation saved Binance – if all assets were in hot wallets, the loss could be billions of dollars
3. The speed of emergency response is very important – it only took a few minutes from the attack to the discovery, and the loss was stopped quickly.

Case 2: \$625 Million Stolen from Ronin Bridge in 2022 – “Social Engineering by North Korean Hackers”

Background: Ronin Bridge is a cross-chain bridge for Axie Infinity (a blockchain game), managed with 9 validator nodes (requiring 5/9 multi-signatures to execute transactions). Who is the attacker? North Korea' s Lazarus group – a state-sponsored hacking group that has launched numerous attacks on the crypto industry. Attack process: Step 1: Social engineering The attacker contacted a senior engineer at Axie Infinity (a game owned by Ronin's parent company Sky Mavis) through LinkedIn. Using "high-paying job opportunities" as bait, he sent a malicious PDF file disguised as a salary package. After the engineer opened the file, the attacker planted malware on his device and gained internal access.

Step 2: Obtain the private key. Through social engineering, the attacker obtained the private keys of 5 validators (just meeting the threshold requirement of 5/9 multi-signature). Step 3: Launch the attack (March 23, 2022). The attacker signed two transactions with 5 private keys and withdrew 173,600 ETH + 25.5M USDC from Ronin Bridge, with a total value of approximately US\$625 million.

Step 4: Delay in discovery (6 full days) The Ronin team did not discover that the funds had been stolen until 6 days later (March 29) when users reported being unable to withdraw money. Why did it take 6 days to discover? Because the daily monitoring frequency of Bridge is not high enough – no one checks "whether the money in Bridge is still there" every day. Core lessons:

1. The "dispersion" of multi-signature is more important than the "quantity" – 5/9 seems safe, but if the holders of 5 private keys are all affected by the same company, it is actually equivalent to a "single point of failure"
2. Social engineering is the most difficult attack to defend against – no matter how powerful the security system is, it can still be broken by "humans"
3. The monitoring frequency must be high enough – it took 6 days to detect anomalies, indicating serious flaws in the monitoring mechanism

Practitioner note: The essence of social engineering attacks is to "exploit human weaknesses" – greed (the temptation of high salary), fear (pretending to be urgent), and trust (pretending to be authority). The key to defense is not technology, but security awareness training.

Case 3: Wormhole's \$320 Million Exploit in 2022 – Signature Verification Bypass

Background: Wormhole is a cross-chain bridge between Solana and Ethereum – users can "bridge" assets from one chain to another. What is the vulnerability? Where is the vulnerability: Signature verification logic in smart contracts. When verifying cross-chain messages, the contract needs to check whether the signature comes from a legitimate "Guardian"

Under normal circumstances, the cross-chain bridge needs to confirm that a group of legal Guardians have signed the cross-chain message before allowing the corresponding assets to be minted on the target chain. The core problem of the Wormhole incident is that the attacker used old logic and verification bypass paths in the signature verification process to allow a message to pass inspection without valid authorization.

Result: The attacker minted 120,000 wETH on the Solana chain, worth approximately \$320 million. The focus of this case is not that a certain line of code is wrongly written, but that problems in any part of the cross-chain system upgrade, old function compatibility, signature verification, and audit processes may cause huge losses. How did you deal with it afterwards?

- Jump Trading (Wormhole's parent company) covered \$320 million in losses out of its own pocket
- This is also one of the largest "out of pocket" compensation in DeFi history

Lessons:

1. Smart contract audit cannot be a mere formality – every code upgrade needs to be re-audited
2. Cross-chain bridge is the most vulnerable link in DeFi – because it involves state synchronization of two chains, it is extremely complex.
3. "Small bugs" may cause "big losses" – In traditional software, a bug may just cause the user interface to display an error; in DeFi, a bug may cause hundreds of millions of dollars in losses

3.3 Proof of Reserves: “Is Your Money Really There?”

Why Proof of Reserves Matters

After the collapse of FTX in 2022, the market no longer only accepts exchanges' unilateral claims that “user assets are safe”, but requires more verifiable on-chain or audit evidence. The reserve certificate is the exchange's proof to the public: “I do have all the assets that users have with me.”

How It Works

Imagine you deposit 1 BTC on an exchange. The exchange needs to prove that it actually holds at least 1 BTC (corresponding to your deposit)

Method: Use Merkle Tree (a data structure)

The exchange packages the balances of all users into a Merkle Tree. After simplification, it can be understood as follows:

Hierarchy	Sample content	function
root hash	Root Hash	Open to the public and used to represent the summary status of the entire tree
intermediate node	Hash(AB)、Hash(CD)	Summarize the balances of multiple users layer by layer
leaf node	Hash(A)、Hash(B)、Hash(C)、Hash(D)	Corresponding to a single user balance record
User balance	Zhang San 1 BTC, Li Si 2 BTC, Wang Wu 0.5 BTC, Zhao Liu 3 BTC	Users can verify whether their balance is included

Zhang San can verify whether his balance is included in the tree. The exchange publicly proves: the assets on the chain $\geq$ the sum of all balances in the tree. The bank publishes a "vault photo" every month to prove that the money is there. But the problem is - the three major limitations of proof of reserves: It can only prove "a certain moment" - just like a photo can only prove that there was money in the vault at the moment when the photo was taken. The bank may have borrowed the money before taking the photo to cover the appearance, and then returned it after taking the photo. It cannot prove that the liability side is complete - the exchange may have concealed some user liabilities. It may "borrow currency to top up" - after the collapse of FTX in 2022, some exchanges were found to temporarily transfer a large amount of assets from other platforms to take snapshots, and then transfer them back after the snapshot. Practitioner note: Reserve certification is only a "first step" and is not a substitute for third-party auditing. Truly credible proof of reserves = consecutive multi-day snapshots + third-party audits + publicly available addresses on the chain. Summary of this chapter

- Exchange security threats come from six dimensions. Real cases prove that even top exchanges cannot escape attacks.
- Proof of Reserve (PoR) provides transparency, but suffers from structural blind spots - it is not a sufficient condition for security commitments
- Security investment is not a cost, but a prerequisite for the continued operation of the exchange.

Chapter 4 Compliance and Licensing

Chapter Overview

Compliance is not a layer of packaging for the exchange, but an access condition that determines whether the business can continue to operate. This chapter introduces the major regulatory frameworks in Mainland China, Hong Kong, the United States, and the European Union, as well as the daily compliance work of exchanges such as KYC, AML, and sanctions screening.

The regulatory information involved in this chapter has been verified based on public information until May 2026. Regulatory policies change rapidly, and formal decisions should still be based on the opinions of local lawyers and the latest announcements from regulatory agencies.

4.1 Why Compliance Matters

The most direct reason: Failure to comply may result in fines, closure, or prosecution. Case: In 2023, Binance reached a settlement of approximately US\$4.3 billion with the U.S. Department of Justice (DOJ) and other agencies, admitting to violating the Bank Secrecy Act, unregistered remittance business, and sanctions-related obligations. This case illustrates that even if an exchange is registered overseas, it may still be subject to the jurisdiction of U.S. law enforcement agencies as long as it serves U.S. customers, uses the U.S. financial system, or violates U.S. sanctions. The deeper reason: Compliance is a long-term business capability. When regulation tightens, it is easier for platforms with licenses, complete internal controls, and clear segregation of user assets to continue operating; platforms that rely on regulatory arbitrage may face fines, market withdrawal, management criminal liability, and bank channel interruption.

4.2 Major Global Regulatory Frameworks

Mainland China: Virtual Currency Trading Services Are Explicitly Prohibited

Key timeline: December 2013, the People's Bank of China and five other ministries and commissions issued the "Notice on Preventing Bitcoin Risks" Qualitative: Bitcoin is not a currency, but a "virtual commodity". Financial institutions are not allowed to carry out Bitcoin-related businesses.

In September 2017, seven ministries and commissions jointly announced: ICO (token issuance financing) was prohibited, domestic crypto exchanges were closed, and exchanges such as Huobi and OKEx were forced to "go overseas"

In May 2021, the 51st meeting of the Financial Stability and Development Committee of the State Council (Financial Committee) named "Cracking down on Bitcoin mining and trading activities"

In September 2021, ten ministries and commissions jointly issued the "Notice on Further Preventing and Dealing with Speculation Risks in Virtual Currency Transactions" - often referred to as the "924 Notice" in the industry. Core points: - Virtual currency-related businesses are "illegal financial activities" - Virtual currencies do not have the status of legal tender - The provision of services by overseas exchanges to residents in China through the Internet is also an illegal financial activity - Participation in virtual currency investment transactions is not legally protected - If any legal person, unincorporated organization, or natural person invests in virtual currencies and related derivatives that violates public order and good customs, the relevant civil legal actions may be deemed invalid, and the resulting losses shall be borne by them themselves.

What This Means for Chinese Practitioners

Risk warning: The exchange operating models, license applications and growth strategies discussed in this handbook are all based on overseas legal frameworks. In mainland China, activities such as virtual currency transaction matching, market making, diversion, promotion, payment and settlement, and technical support for domestic residents may trigger regulatory red lines and even criminal risks. Even if they are outside the country, mainland practitioners should carefully evaluate service targets, contract entities, capital flows, publicity channels and team locations.

Hong Kong, China: Licensed Operations with Strict Investor Protection

Since 2023, Hong Kong, China, has established a relatively complete licensing system for virtual asset trading platforms. Special explanation is required: The Hong Kong system is aimed at platforms operating in Hong Kong or actively promoting services to Hong Kong investors, and does not constitute a channel for serving residents of mainland China. Key points of Hong Kong's VASP licensing system: In June 2023, the Hong Kong SFC (Securities and Futures Commission) officially implemented the virtual asset service provider licensing system. Core requirements: - All centralized exchanges operating in Hong Kong must be licensed - Licensed platforms can provide services to retail investors after meeting suitability, risk disclosure and token inclusion standards - Exchanges must meet capital adequacy requirements (not less than HK\$5 million) - Compensation arrangements or insurance arrangements must be established for customer virtual assets - Client assets must be stored segregated from the exchange's own assets - subject to AML/CFT compliance requirements

As of the public list updated on the SFC official website on February 16, 2026, Hong Kong SFC has officially licensed a number of virtual asset trading platform operating entities,

including OSL Exchange, HashKey Exchange, HKVAX, HKbitEX, Accumulus, DFX Labs, EX. IO, PantherTrade, YAX, Bullish, BGE, VDX, etc. The list is subject to continuous change, and by publishing the list, SFC does not guarantee the performance or credibility of the platform. Before actual cooperation, account opening or citation, the "List of Licensed Virtual Asset Trading Platforms" on the SFC official website should prevail.

Why Hong Kong Matters to Mainland Practitioners

- Compliance status: Licensed to operate in Hong Kong and can serve Asian users within the legal framework
- Fiat channel: The Hong Kong dollar is pegged to the U.S. dollar, and fiat deposits and withdrawals are relatively convenient.
- Talent reserve: A large number of Chinese blockchain practitioners gather in Hong Kong
- The rules are relatively clear: the requirements for licenses, customer asset custody, token listing review, anti-money laundering, customer support and complaint handling, etc. are relatively clear. However, please note: a Hong Kong license does not equal a authorization to serve mainland China residents. There are still significant legal risks in providing trading services to mainland residents. In practice, it is necessary to strictly implement user geographical isolation, IP and device identification, marketing restrictions, KYC residence verification and employee behavior management.

United States: Overlapping Federal and State Regulation

The United States does not have a unified encryption regulator, but multiple federal agencies plus state regulation in parallel. It has always been the most difficult market for compliance in the world.

Multiple regulations at the federal level:

SEC (Securities and Exchange Commission) – Regulates crypto-assets recognized as "securities" and related trading, brokerage, and clearing activities – Core dispute: Which tokens or trading arrangements are securities? (The Howey test remains an important judgment framework)

CFTC (Commodity Futures Trading Commission) – manages crypto asset derivatives, futures, swaps and related market manipulation – BTC, ETH, etc. are treated as commodities in multiple law enforcement and regulatory scenarios, but specific businesses still need to be judged on a case-by-case basis

FinCEN (Financial Crimes Enforcement Bureau) – Anti-Money Laundering Compliance – Businesses that provide services such as convertible virtual currency exchange, transmission, etc. are generally required to register as an MSB (Money Services Business) and meet AML obligations

State Regulation – Each state has independent licensing requirements – New York's BitLicense is one of the most stringent state licenses in the United States, and the application cycle

may be as long as 2-3 years. The vast majority of exchanges initially choose to bypass New York users

Key timeline:

In 2013, Coinbase completed MSB registration with FinCEN and was one of the earliest compliant exchanges. In 2015, BitLicense was launched in New York, sparking industry controversy. In 2021, Coinbase was directly listed on Nasdaq and was the first major crypto exchange to be listed in the United States. In 2023, Binance reached a \$4.3 billion settlement with the DOJ (the largest fine in industry history). In 2024, the regulatory tug-of-war between the SEC and multiple exchanges continued, and the regulatory framework remained unclear. In 2025, the SEC established the Crypto Task Force to try to promote a clearer regulatory framework for crypto assets. In 2025, US regulatory and legislative discussions began to turn more to market structure, stablecoins, custody and disclosure rules.

Lessons for practitioners: When operating a crypto business in the United States, you cannot just ask "do you have an exchange license?" Securities laws, commodity derivatives rules, FinCEN MSB registration, state-level money transfer licenses or BitLicenses, OFAC sanctions compliance, consumer protection, tax and data privacy requirements often need to be evaluated simultaneously. The U.S. market has a large amount of capital, but compliance costs and enforcement risks are equally high.

The following uses the real paths of two leading exchanges to show the costs and returns of the two strategies:

Two Compliance Paths: Coinbase vs Binance

Coinbase – "Compliance First" Coinbase has made "compliance" a core strategy from the first day of its establishment in 2012: Obtained MSB registration from FinCEN in 2013 Obtained New York State BitLicense in 2015 (BitLicense is known for its strictness, with an application cycle of 2-3 years) Coinbase was one of the first exchanges to obtain a BitLicense Directly listed on Nasdaq in 2021 (not IPO) The market capitalization exceeded US\$85 billion on the first day of listing. 2023. Obtain Singapore MPI license. 2024-2025. Promote CASP authorization layout under the EU MiCA framework. Cost: slower development – compliance requires a lot of time and money investment. Return: Although Coinbase faced lawsuits and regulatory pressure from SEC and other agencies when regulations were tightened, its ability to withstand pressure was significantly higher than that of most offshore exchanges due to its relatively complete foundations of disclosure, auditing, banking channels, and corporate governance. **Binance – the "do it first" school** Binance has developed extremely fast, but has taken detours in compliance: In 2017, it was launched by the Chinese-Canadian Changpeng Zhao (CZ) team and registered in the Cayman Islands. Most of the core team members work in China. In 2018, it moved again to Malta ("Blockchain Island"). 2019-2020 Operated in a "decentralized headquarters" model. No clear legal entities in multiple jurisdictions. 2021

Regulatory warnings from many countries followed: The British FCA banned Binance from operating in the UK. The Japanese FSA issued a warning for unregistered operations. The US SEC investigation is ongoing. In 2022, the "compliance transformation" began: Hiring a former US Treasury official as chief compliance officer. applying for licenses in France, Italy, Spain and other countries. In 2023, reaching a settlement of approximately US\$4.3 billion with the US DOJ and other institutions. CZ resigned as CEO. Summary: Compliance is not a one-time license, but a set of ongoing operational capabilities. Coinbase's "compliance first" strategy sacrifices some speed in exchange for a stronger foundation of banking, audit and institutional trust. Binance's "do it first, tell it later" strategy helped it expand rapidly, but ultimately came at huge enforcement and governance costs. The experience of the U.S. market shows that when regulations become stricter, compliance investment will change from a cost item to a survival condition.

4.3 KYC and AML: Identity Verification and Anti-Money Laundering

KYC: Know Your Customer

KYC means that the exchange requires you to provide proof of identity to confirm that "you are you". Why is KYC required?

- Prevent money laundering, terrorist financing and other criminal activities
- Meet regulatory requirements (KYC is required by almost all formal exchanges)
- Protect users - If an account is compromised, KYC information can help with recovery.

There is no global template for KYC grading. The following is a common example of productization layering in exchanges. The specific materials, quotas and review standards are subject to the regulatory requirements and platform rules of the platform location:
 Level 1 (Basic Verification) Only requires: email/mobile phone + name + date of birth
 Restriction: Maximum withdrawal of \$2,000 per day Suitable for: small-amount trading users

Level 2 (enhanced verification) Requires: Upload ID card/passport + face recognition + address proof Restriction: Maximum withdrawal of \$50,000 per day Suitable for: Users with medium trading volume

Level 3 (Advanced Verification) Requires: Proof of Address + Proof of Source of Wealth + UBO Information Limit: Withdrawal of \$500,000+ per day Suitable for: High Net Worth Users and Institutions

AML: Anti-Money Laundering

AML is a system used by exchanges to identify and prevent money laundering. What does money laundering look like in the crypto world? Here's an example: A hacker stole 10,000 BTC and wanted to turn it into "clean money." He may:

1. Spread BTC across hundreds of addresses ("split")

2. Use a coin mixing service (such as Tornado Cash) to wash out source information (“coin mixing”)
3. Finally, the funds are sold on an exchange and converted into fiat (“cashed out”). The exchange’ s AML system is designed to intervene before that step by analyzing on-chain fund flows and identifying whether the money may be tainted. Common AML tools include Chainalysis, Elliptic, and TRM Labs. These systems assign risk scores to incoming funds. If the source is linked to a hacker address, fraud address, sanctioned address, or high-risk mixer path, the exchange may freeze the account, request additional materials, or file a suspicious activity report with regulators or law enforcement. Case: Tornado Cash sanctions. In August 2022, OFAC added Tornado Cash-related addresses to the SDN sanctions list because the protocol had been used to launder illicit funds.
 - Many compliant exchanges have blocked addresses that interact with Tornado Cash
 - Circle (USDC issuer) freezes USDC interacting with Tornado Cash
 - GitHub once restricted access to relevant code repositories, and subsequent processing changed with adjustments to laws and platform policies.
 - The controversy caused by the arrest of the Tornado Cash developer in the Netherlands: Some users who have not committed any direct illegal acts may also be affected. For example, attackers send small amounts of ETH to random addresses ("dust attacks"), causing these addresses to become on-chain associated with high-risk protocols. Therefore, compliant exchanges need to distinguish between three types of situations: "direct use of high-risk services", "passive receipt of contaminated funds" and "unexplained sources of funds" to avoid accidental injuries caused by mechanical bans.

4.4 Regulatory Arbitrage and Its Risk Boundaries

What is regulatory arbitrage? Regulatory Arbitrage refers to companies taking advantage of differences in regulatory standards, enforcement intensity, and license costs in different jurisdictions to split registration entities, teams, servers, fiat channels, and user services into different regions. In the early days of the crypto industry, this practice was common; however, as countries strengthen cross-border law enforcement, bank due diligence, and sanctions compliance, it has become increasingly difficult to reduce substantial risks by simply relying on "registration in offshore locations."

This section is not to provide operational paths to avoid regulation, but to help readers identify compliance risks in exchange operations. In practice, to judge whether a platform is compliant, we should not just look at the place of registration or license name, but also which users it actually serves, where the funds come in and out, where the marketing is targeted, and where the team and actual controller are.

Common Regulatory Arbitrage Tactics

① “Headquarters” in a place with loose regulation Common choices: - Seychelles - almost no crypto regulation - Cayman Islands - traditional offshore financial center - Malta - once known as the "blockchain island" - Dubai - emerging crypto-friendly center

Actual operations: - Teams are located in Singapore, London, San Francisco and other places - Servers are distributed around the world - Users are spread all over the world

Problem: The nominal "headquarters" is just a shell company. The real operations team is in another country. Users don't know who to call when they have problems.

② "License" is in a place with low regulatory requirements. Strategy: First obtain a license in a place with low regulatory requirements, and then claim to be a "licensed exchange" but may not actually obtain a license in the market where it actually operates.

Example: An exchange claims to "hold a license from country X" but the license requirements of country

③ Isolate risks through “subsidiaries” Strategy: Establish independent subsidiaries in compliance markets (such as the United States). Subsidiaries strictly comply with regulations and operate independently, but the parent company (global version) may have lower compliance standards

Example: FTX has FTX US (U.S. version) and FTX (international version). FTX US claims to operate independently and has strict compliance, but actually shares resources with the international version. After FTX collapsed, FTX US was also affected.

Consequences of Regulatory Arbitrage

Practitioner note: As a user, what you need to pay attention to is not "where the exchange is registered", but "whether it has a compliance license in the country where you are located." If the exchange is not licensed in your country, you may not be able to pursue your rights through local laws if something goes wrong.

Daily compliance work of exchanges usually includes:

Compliance module	Daily operational actions
KYC/KYB	Verify personal identity, institutional entity, ultimate beneficiary and source of funds
AML transaction monitoring	Identify abnormal deposits, split transfers, high-risk addresses and suspicious transaction patterns
Sanctions Screening	Continuous screening of OFAC, UN, EU and local sanctions lists

Compliance module	Daily operational actions
Travel Rule	Transfer and save information on both parties to the transfer in eligible jurisdictions
market behavior monitoring	Monitor the risks of blackmail, spoofing, layered orders and insider trading
Customer asset isolation	Distinguish between the platform's own assets and customer assets, and cooperate with auditing, PoR and regulatory reporting
Complaints and record retention	Save KYC, transaction, customer support, risk management and manual approval records to meet regulatory inspections

Chapter Summary

- Global encryption regulation remains fragmented, with mainland China's ban, Hong Kong's VASP licensing, US multi-regulatory regulation and the EU's MiCA framework representing several typical paths
- Compliance is changing from an option to a barrier to entry. The focus is not just on licenses, but also on customer asset segregation, AML, sanctions screening, market monitoring and customer support and complaint handling.
- The window period for regulatory arbitrage continues to narrow. In the long run, compliance and governance capabilities, together with technology and liquidity, will constitute the core competitiveness of the exchange.

Chapter 5 User Growth and Operations Strategy

Chapter Overview

Exchange growth comes from the mutual reinforcement of users, trading volume and liquidity. This chapter discusses how exchanges acquire, retain, and promote customers, and use an indicator system to determine whether growth is healthy.

5.1 Growth Flywheel: How the Snowball Gets Bigger

The growth of exchanges follows a "flywheel effect":

1. More users bring more trading volume.
2. More trading volume improves market depth and price quality.
3. Better liquidity reduces transaction costs.
4. Better prices and experiences continue to attract new users.

How to start this flywheel? Each link has a corresponding strategy:

5.2 Case Study: Comparing Two Growth Strategies

Strategy 1: Crypto.com's Cash-Burning Growth

Crypto.com's strategy focuses on high-intensity branding and sponsorship to acquire customers. approximately 1 million users in 2019 Marketing expenditures peaked in 2021: - US\$700 million to buy the naming rights of the Staples Center in Los Angeles (20 years) - Sponsoring F1 racing (5-year global cooperation agreement, total price of approximately US\$100 million) - Sponsoring the UFC (US\$175 million/10 years) - Sponsoring the World Cup (rumored to be hundreds of millions of US dollars) - Matt Damon endorsement advertising The number of users is growing rapidly, breaking through in early 2022 50 million registered users in 2022. The bear market is coming: - 20%-25% layoffs - Reduce user rewards - User growth stagnates. What is the effect?

- Industry analysts estimate that the user acquisition cost (CAC) is about \$70-\$100/user
- Compare Binance's CAC: about \$10-\$20/user (mainly relying on word-of-mouth and referral commissions)
- A large number of users are short-term incentive-driven - they come for registration rewards and will no longer be active after the rewards are over.

Lesson: Burning money can quickly acquire customers, but it cannot guarantee retention. A true growth flywheel requires the traction of the product itself.

Strategy 2: Binance's Referral Commission System

Binance's referral commission mechanism is the most successful growth engine in the industry:

Mechanism: The referrer receives a 20% commission on the referral's trading fee. If the referrer holds ≥ 500 BNB, the commission is increased to 40%. The referred person also receives a 10%-20% commission discount.

Example: A KOL recommended 10,000 users. The average monthly trading volume per person was \$5,000. The trading fee was 0.1%.

Monthly trading fee = $10,000 \times \$5,000 \times 0.1\% = \$50,000$ KOL monthly rebate = $\$50,000 \times 40\% = \$20,000$ - constituting considerable passive income Why is it so effective?

- For recommenders: Passive income, the more referrals you make, the more you earn
- For referrals: Direct fee discount
- For Binance: Customer acquisition cost is controllable (only paid when actual transactions are generated)
- Flywheel effect: the more recommenders $\rightarrow$ the more users $\rightarrow$ the better the liquidity $\rightarrow$ more people come $\rightarrow$ more people recommend

5.3 Growth Funnel: Where Users Drop Off

1 million people see your ad $\rightarrow$ 2-5 million people click (CTR 2-5%) $\rightarrow$ 2,000-15,000 people register (registration rate 10-30%) $\rightarrow$ 800-10,500 people complete KYC (KYC rate 40-70%) $\rightarrow$ 240-5,250 people deposit for the first time (first deposit rate 30-50%) $\rightarrow$ 144-4,200 people complete their first transaction (first transaction rate 60-80%) $\rightarrow$ 29-1,680 people are still trading after 30 days (retention rate 20-40%)

Estimated based on the industry median: 1 million impressions $\rightarrow$ 35,000 clicks $\rightarrow$ 5,000 registrations $\rightarrow$ 2,800 KYC $\rightarrow$ 1,100 first deposit $\rightarrow$ 770 first transaction $\rightarrow$ 230 retention. That is, for every 1 million ad impressions, about 230 people will eventually become 30-day retained users. The key to optimization: find the layer with the most loss and focus resources on fixing it. If the conversion rate from registration to KYC is particularly low $\rightarrow$ it may be that the KYC process is too cumbersome. If the conversion rate from KYC to first deposit is particularly low $\rightarrow$ it may be that the deposit channel is inconvenient. If the conversion rate from first deposit to first transaction is particularly low $\rightarrow$ it may be that novice guidance is not good enough. How to optimize the registration-transaction conversion rate? Registration $\rightarrow$ Analysis of the churn point of the first transaction:

Loss point one: KYC is too cumbersome Optimization: step-by-step guidance, AI automatic identification, mobile optimization

Loss Point 2: Don't know how to make the first purchase Optimization: novice tutorials, simulated trading, "one-click currency purchase" function

Drop-off point 3: deposits are inconvenient. Optimization: support more fiat deposit channels and P2P options.

Loss point 4: Forgot after registration Optimization: Send reminder email/push within 24 hours after registration

5.4 Community Operations: Turning Users into Distribution Nodes

Common community positions:

- X / Twitter: Suitable for publishing announcements, market opinions, event information and brand volume building
- Telegram: suitable for high-frequency interaction, event operations and local language community management
- Discord: suitable for project-based communities, developer communities, equity stratification and long-term user accumulation
- Content platforms such as YouTube / TikTok / Xiaohongshu: suitable for tutorials, market interpretation and brand education

Community operation methodology:

- AMA (Ask Me Anything): regularly invites founders/executives to have direct conversations with the community
- Community governance: Let users participate in currency voting and product decision-making
- KOL Cooperation: Establish long-term cooperative relationships with industry opinion leaders
- UGC incentives: encourage users to share trading strategies and market analysis

Compliance boundaries: Community operations cannot just pursue transformation. Marketing to restricted areas, promising returns, inducing highly leveraged transactions, and KOL promotions without disclosing interests may all bring compliance and reputational risks.

5.5 Growth Metrics: Measuring the Business with Data

Growth cannot rely on “gut feel”; it has to be managed with data. The following are the core metrics used in exchange operations:

North Star Metrics

What is the North Star Indicator? It's “If you could only look at one number, what would you look at?” - it represents the core value of the business. For exchanges, Trading Volume is the North Star indicator. Because trading volume = user activity × product experience × market conditions, it is the final result of all growth efforts.

Core Growth Metrics Explained

DAU/MAU: user stickiness

Assume that the platform has 1 million monthly active users (MAU = 1 million), of which 300,000 people log in to trade every day (DAU = 300,000).

$$\text{DAU/MAU} = 300,000 / 1 \text{ million} = 30\%$$

This means: 30% of your monthly active users come "every day". The higher the DAU/MAU, the stronger the user stickiness.

CAC: customer acquisition cost

CAC = market launch, rebates, incentives and channel costs / number of new effective users. Looking only at registered users will be inflated. Exchanges should look at the effective CAC after completing KYC, first deposit and first transaction.

LTV: User lifetime value

LTV = fees, interest, custody fees and other service income contributed by users during the life cycle. Healthy growth is not about "the more registrations, the better", but about LTV covering CAC and leaving enough profit margin.

First deposit rate and first transaction rate

The first deposit rate reflects deposit channels and trust, and the first transaction rate reflects the cost of novice guidance and product understanding. The problem with many exchanges is not that there is no traffic, but that users do not know how to safely complete their first transaction after registering.

Net deposits and deposits

Net deposit = deposit - withdrawal. It can better reflect whether users actually keep their assets on the platform than pure trading volume. The 30-day and 90-day retention rates are used to determine whether users have formed long-term usage habits.

Chapter Summary

- The exchange's user growth is essentially a "flywheel" rather than a "funnel" - trading volume, liquidity and user number reinforce each other
- The ratio of customer acquisition cost (CAC) to user lifetime value (LTV) is a core indicator of growth health
- Different market stages require different growth strategies: bull market grabs traffic, bear market retains

Chapter 6 Exchange Products and Service Systems

Chapter Overview

Exchanges are more than places to “buy and sell Bitcoin.” The richer the product line, the longer users stay and the more monetization scenarios the platform can create. This chapter summarizes common exchange product lines, institutional services, and liquidity infrastructure.

6.1 Product Line Overview

The exchange's products can be divided into three broad categories. Let's talk about it one by one.

1. Trading Products

1. Spot trading (Spot): the most basic trading

Users use USDT (a stablecoin anchored 1:1 with the U.S. dollar) to directly buy BTC. After the purchase, the assets belong to the user. Profit when the price goes up and loss when the price goes down. There is no leverage and no need to borrow money. Spot trading is suitable for novices, long-term holders and users who do not want to bear the risk of liquidation.

2. Margin: borrow money to buy coins

If a user has 1,000 blocks and wants to buy 3,000 blocks of BTC, the exchange can lend 2,000 blocks. If BTC rises by 10%, the user earns 300 yuan (yield rate is 30%); if it falls by 10%, the user loses 300 yuan (yield rate -30%). The difference from spot trading is that spot trading can only use its own funds. Leverage can amplify profits and losses simultaneously.

Tip: Leveraged funds need to pay interest (annualized is usually in the range of 5%-20%), and when the loss exceeds margin safety boundary, forced liquidation will be triggered.

3. Perpetual Futures: Obtain long and short price exposure

Perpetual contracts do not require users to actually buy BTC, but instead trade contract exposure for the rise and fall of BTC prices.

- Go long: Open a "long BTC" contract. If BTC rises from \$60,000 to \$66,000 (a 10% increase), the contract profit will be 10%; if 10x leverage is used, the return on initial capital will be approximately 100%.
- Short selling: Open a "short BTC" contract. If BTC falls from \$60,000 to \$54,000 (a 10% fall), the contract will make a profit of 10%.

Perpetual contracts have no expiration date and do not have to be settled at maturity like traditional futures; however, users need to pay or receive funding rates regularly (see

Chapter 7 for details). Futures trading is usually one of the most profitable products on exchanges because leverage magnifies the nominal trading volume; at the same time, it is also one of the products where ordinary users lose the most money.

4. Options: Buy a right

The core of options is "rights" rather than "obligations". Users pay a premium for the right to buy or sell BTC at a certain price in the future. If the market is unfavorable, you can choose not to exercise the option, and the maximum loss is usually the premium.

Use insurance to understand:

- Buying a put option can be understood as buying a downside insurance for your BTC position.
- If BTC does fall below the agreed price, options bring compensation or hedging benefits.
- If BTC does not fall, the premium is equivalent to an insurance premium.

Options are suitable for institutional users who want to hedge risks, and for traders who are willing to use small premiums to obtain non-linear returns.

5. Grid Trading: Automatically buy low and sell high

Users set a grid within a certain price range, such as \$55,000–\$65,000 for BTC. The program will automatically buy in the low price range and sell in the high price range to earn the price difference in the volatile market.

Example:

- Set up 10 grids between \$55,000–\$65,000.
- BTC fell to \$58,000, and the program automatically bought.
- BTC rises to \$61,000 and the program automatically sells.
- Each time the range price difference is earned, grid income is formed after deducting the trading fee.

Grid trading is suitable for volatile market conditions. If the market rises or falls unilaterally, the grid strategy may underperform the spot holdings, or even suffer larger losses.

6. Copy Trading: Copying other people' s trading operations

The user selects a trader as the follower, and the system automatically copies the other party's trading operations. For novices, copying lowers the decision-making threshold; for platforms, copying products can improve user retention and futures trading activity.

Tips:

- Historical performance is not indicative of future performance.

- Some "experts" may just have good luck in the short term and do not have stable trading capabilities.
- Copying usually requires additional profit sharing, and 10%-30% of the profit may be distributed to the traders being followed.

2. Value-Added Products: Increasing User Retention

① Earn products – users deposit crypto with the exchange to earn yield, similar in user experience to a bank deposit or Yu' e Bao. The exchange may deploy those assets for lending or staking and share part of the yield with users. Flexible products can be withdrawn at any time and usually offer lower rates, often around 1%-5% annualized. Fixed-term products lock assets for a set period, such as 7, 30, or 90 days, and usually offer higher rates, often around 5%-15% annualized. Tip: not all Earn products are safe. High-yield products on smaller exchanges may operate like Ponzi schemes, using later users' funds to pay earlier users' interest.

② Staking – Locking helps maintain the network. Some cryptocurrencies (such as ETH, SOL) adopt a "proof of equity" mechanism – you lock the currency to help verify transactions, and the network rewards you with interest, similar to a time deposit. How does the exchange help you operate: You stake your ETH on the exchange, and the exchange helps you run the verification node, obtains rewards from the network, and distributes them to you (the exchange usually charges a service fee of 10%-25%).

③ Launchpad / IEO – the initial subscription of new coins, similar to "new stocks". If a new project wants to issue tokens, it chooses to be launched on an exchange. Exchange users can use platform coins (such as BNB) to participate in subscriptions and buy new tokens at the issue price. Case: Binance Launchpad' s early launch project BitTorrent Token (BTT) debuted on Binance Launchpad in January 2019 Issue price: \$0.00012 Subscription time: \$7.2 million in financing completed within 15 minutes

After the launch, the price of BTT skyrocketed to \$0.0012 (approximately 10x). Users who participated in the subscription earned 10x in 15 minutes.

Follow-up projects: Multiple Launchpad projects have increased by 5-20x after being launched. Binance Launchpad has therefore received extremely high market attention. Tips:

- Not all Launchpad projects can rise – many projects fall below the issue price after going online
- You need to hold platform currency to participate (for example, Binance Launchpad requires holding BNB)
- In essence, it is "creating new products in the primary market", and the risk is higher than that of secondary market transactions.

④ OTC / P2P trading – buying and selling crypto directly with counterparties. OTC (over-the-counter) trading does not go through the exchange order book; the trade is negotiated directly with a counterparty and is often used for large orders, such as trades above \$50,000, to avoid order-book slippage. P2P (peer-to-peer) trading lets buyers and sellers trade directly while the exchange acts as an escrow platform. Some overseas platforms support bank cards or local payment methods. In mainland China, however, virtual-currency-related trading activities are not legally protected, and using bank cards or third-party payment tools for such transactions may lead to frozen accounts, tainted funds, and compliance risk. Key P2P risks:

- Receiving tainted funds (the most serious risk): If the source of the counterparty's funds involves telecom fraud, gambling, money laundering or other crimes, the receiving account may be frozen or even investigated. Prevention suggestions: Avoid large amounts and transactions during abnormal periods, keep complete transaction records and communication screenshots, and carefully assess local legal risks.
- The counterparty loses contact or refuses to cooperate: Although the exchange has a guarantee mechanism, disputes may still arise
- Price opacity: P2P prices are typically 1%-3% higher than exchange spot prices

3. Institutional Services: The VIP Lounge

① OTC Bulk Trading Desk If you want to buy \$5 million of BTC at one time, buying directly on the exchange order book will cause a lot of slippage (the sell order will be eaten through, and the price will get higher the more you buy). The OTC bulk trading desk helps you find the counterparty and complete the transaction at a one-time negotiated price without affecting the market price.

② Custody helps institutional customers keep their assets, similar to a bank safe. Institutional customers (such as funds and listed companies) are usually unwilling to manage their own private keys – because if the private key is lost, it is really gone. Escrow services provide professional secure storage, insurance and auditing.

③ Prime Brokerage is a one-stop institutional trading service that aggregates liquidity from multiple exchanges, allowing institutions to trade on different exchanges on one platform. Similar to "brokerage dealers."

④ API service: Program interface for quantitative trading team. The quantitative team obtains real-time market conditions, places orders, checks accounts, etc. through APIs to implement programmed automatic trading.

6.2 Market Makers: Liquidity Providers for Exchanges

Market Maker is a very important but often overlooked role in the exchange ecosystem.

What Does a Market Maker Do?

Simply put, market makers are professional traders who continuously provide bid and ask quotes in the market, mainly earning bid-ask spreads and exchange incentives. The market maker's operation: List simultaneously on the BTC/USDT trading pair: Buy order: \$59,950 to buy 1 BTC Sell order: \$60,050 to sell 1 BTC

$$\text{Spread} = \$60,050 - \$59,950 = \$100$$

If both orders are filled: The market maker earns \$100 (price difference income). The net profit after deducting trading fees may be \$60-\$80.

Why Exchanges Need Market Makers

Core reason: Market makers provide "liquidity" to the market - allowing both buyers and sellers to complete transactions quickly. Market without market makers: Buy one for \$59,000 0.5 BTC Sell one for \$61,000 0.3 BTC Spread: \$2,000 (3.3%) ← Huge spread and poor trading experience Market with market makers: Buy one for \$59,950 5 BTC Sell one for \$60,050 5 BTC Spread: \$100 (0.17%) ← Small spread, good trading experience

Types of Market Makers

External professional market maker. Independent quantitative institutions or professional trading companies sign market-making agreements with exchanges to provide liquidity based on indicators such as price spread, depth, and online time.

The project team appoints a market maker. For newly listed tokens, project teams often hire market makers to support the initial order book. The goal is to avoid thin depth and sharp price jumps immediately after listing.

The exchange's self-operated or associated market-making team. This type of model requires a high degree of caution. Information isolation, account identification, fund isolation and conflict of interest disclosure mechanisms must be established, and user assets cannot be used to make markets.

Cross-market arbitrage market maker. Connect multiple exchanges at the same time and use price differences to level prices on different platforms, which also helps the market form a more consistent price.

Market-Maker Income and Risk

Source of income:

- Bid-ask spread (main income)
- Exchange commission rebate (the exchange provides market makers with preferential Maker rates, or even negative rates - market makers can make money by placing orders)

Risks:

- **Inventory risk:** Market makers hold both BTC and USDT. If BTC plummets, the BTC held by the market maker will depreciate significantly.
- **Adverse selection risk:** Informed traders (such as insiders) may trade without the knowledge of the market maker, causing the market maker to lose money
- **Systemic risk:** A bug in the market maker's algorithm may result in huge losses within seconds.

Practitioner note: Alameda Research (an affiliate of FTX) was once one of the largest market makers in the crypto industry. After the collapse of FTX, it was discovered that Alameda not only made markets, but also misappropriated FTX user funds. The close relationship between market makers and exchanges was one of the important reasons for the collapse of FTX.

6.3 VIP Fee Tiers: The More You Trade, the Lower the Fee

Level	30-day trading volume	Maker rates	Taker rate	carried interest
Ordinary user	< \$100K	0.10%	0.10%	Basic functions
VIP 1	≥ \$100K	0.08%	0.10%	Dedicated customer support
VIP 2	≥ \$500K	0.06%	0.08%	Higher API limits
VIP 3	≥ \$5M	0.04%	0.06%	Exclusive event invitations
VIP 4	≥ \$20M	0.02%	0.04%	Dedicated account manager
VIP 5	≥ \$100M	0.00%	0.02%	highest priority

Design logic:

- **Maker < Taker:** Encourage pending orders to provide liquidity
- **Decreasing ladder:** The greater the trading volume, the lower the fee rate, which encourages large investors to concentrate on transactions.
- **Negative Maker rate:** Top exchanges give commissions to large Makers (rate < 0)

Note: The above rates are schematic data and are only used to illustrate the tiered pricing logic. The actual rates and grading standards of each exchange are subject to the announcement on its official website.

6.4 Customer Support: Who Helps Users When Something Goes Wrong?

Customer support is the first-line user contact point of the exchange – when users encounter problems, the quality of customer support directly determines whether the user stays or leaves.

Tiered Customer Support Structure

Frontline Customer Service (L1) – "Front Desk Reception" – Handles 80%+ of common questions – "How do I deposit?" – "Why hasn't the withdrawal arrived yet?" – "How can I retrieve my forgotten password?" – "Why has the KYC review not been passed?" – 7×24 multi-language online support – Work order system + online chat + phone – Use FAQ and knowledge base to solve most problems

Second-line customer support (L2) – "Technical experts" – Handle complex upgrade issues – "Why was my transaction canceled?" – "Why are my assets so low?" – "How to solve API interface errors?" – 7×24 technical support – Priority processing for VIP users

Third-line customer support (L3/expert team) – "Crisis handling" – Handling the most difficult issues – Security incident response (account theft) – Regulatory complaint handling – Large asset disputes – Usually coordinated by the security team and compliance team

SLA: How Quickly Must Support Respond?

SLA (Service Level Agreement) is the platform's internal commitment to response speed, processing time limit, and upgrade path. Different platforms have different standards. The following table is only a training example:

Customer type	first response target	solve goals	Upgrade conditions
Ordinary user	5-15 minutes	24-48 hours	Complaints involving funds, account security, and supervision
VIP user	1-5 minutes	4-24 hours	Large withdrawals, liquidation disputes, API failures
Institutional clients	Exclusive channel	Contractual SLA	Abnormalities in custody, clearing, market making, and fiat channels

Lessons for reference: Customer support is the “hidden competitiveness” of the exchange. AI customer support is replacing a large number of standard questions and answers, but issues

involving account security, compliance review, large assets, liquidation disputes and institutional customers still require the intervention of second- and third-line experts.

When users encounter problems, if they can get an effective response within a few minutes, it is easier to continue to trust the platform; if there is no feedback for a long time, the risk of churn and complaints will increase. Operational tip: If users encounter problems on the exchange, they should usually give priority to using online chat and retain the ticket number. Online chat is more suitable for immediate troubleshooting, and work orders are more suitable for leaving traces of complex problems; VIP or institutional users should be handled through a dedicated account manager and the SLA path agreed in the contract. Summary of this chapter

- The exchange product line is divided into three layers: trading, finance and infrastructure. The standard for high-quality products is to reduce user decision-making costs.
- Prediction markets are an industry-extension topic. Readers should read Chapter 16 after they have mastered the main exchange operating model.
- Market makers are not counterparties, but infrastructure providers of the trading ecosystem.
- The core tension of product design: making professional users feel powerful and making ordinary users feel simple enough

Chapter 7 Risk Management and Anti-Manipulation

Chapter Overview

This chapter discusses the riskiest and most controversial aspects of exchanges: leveraged trading and clearing. Through specific numerical cases, it explains how leverage magnifies gains and losses, and why the liquidation mechanism affects user experience and platform risks.

7.1 Leverage and Liquidation Risk

Risk warning: Leveraged trading and futures trading may result in a total loss of capital, or even a account deficit after liquidation (loss exceeding margin) under extreme market conditions. All cases below are only to help understand the mechanism and do not constitute any trading advice. Please participate with caution after fully understanding the risks.

What Is Leverage?

In short, suppose you have 100 yuan but want to buy something worth 1,000 yuan. You borrow the remaining 900 yuan. Initial capital: \$1,000 (margin). Leverage: 10x. Actual trade size: $\$1,000 \times 10 = \$10,000$.

Exchanges usually set a "maintenance margin rate" (such as 0.5%), which triggers liquidation when your margin balance falls below 0.5% of the position value. Different exchanges and different currencies have different maintenance margin rates.

You bought BTC for \$10,000

If BTC rises by 10%: Your profit = $\$10,000 \times 10\% = \$1,000$ Yield = $\$1,000 / \$1,000 = 100\%$ - Double your initial capital

If BTC falls by 10%: Your loss = $\$10,000 \times 10\% = \$1,000$ Loss rate = $\$1,000 / \$1,000 = 100\%$ - all initial capital is lost Practitioner note: Leverage is a "double-edged sword" - when it goes up, your profit doubles, and when it goes down, your loss doubles. Many people have only seen that "10x leverage will double if it rises by 10%", but they did not expect that "10x leverage will clear if it falls by 10%".

What Is Liquidation?

When a user's loss approaches the margin safety margin, the exchange will forcefully close the position - this is "liquidation" or "liquidation."

Why force liquidation? The core purpose is not to simply allow the exchange to "not lose money", but to close risk exposures before margins are insufficient to prevent losses from penetrating to insurance funds, counterparties or the platform itself.

A Full Liquidation Example

Zhang San's operation: Margin: \$1,000 Leverage: 10x Operation: Long BTC (bet on the increase)
Entry price: \$40,000 Position: 0.25 BTC (= \$10,000 / \$40,000)

Price change process:

\$40,000 → \$38,000 (down 5%) Loss = $0.25 \times (\$40,000 - \$38,000) = \$500$ Remaining margin =
\$1,000 - \$500 = \$500 Status: Still holding up

\$38,000 → \$36,000, or 10% below the entry price. Loss = $0.25 \times (\$40,000 - \$36,000) = \$1,000$.
Remaining margin = \$1,000 - \$1,000 = \$0. Status: the margin is fully depleted and liquidation is triggered. What does the exchange's liquidation engine do? Different exchanges handle this differently. Method 1: staged liquidation, which is more user-friendly. The engine first closes part of the position, such as 50%, to reduce risk. If the remaining position is still too risky, it closes another portion, giving the user more breathing room. Method 2: full liquidation, which is safer for the exchange. Once liquidation is triggered, the entire position is closed, the user loses the full margin, and the exchange reduces the chance of bearing deficit risk.

Note: The above case is a simplified demonstration, with "margin returning to zero" as the trigger point. In fact, the exchange has a "maintenance margin rate" (the 0.5% mentioned above), which will intervene in forced liquidation when the book loss is close to but the margin has not been completely exhausted. The purpose is to prevent the loss of the position from being passed on to the platform. The actual liquidation price was slightly higher than the \$36,000 in the case.

Insurance Fund

When a user is cashed out, a "negative equity" sometimes occurs – the loss exceeds margin, resulting in a negative balance. How did the overflow occur? In extreme market conditions (such as a 30% plunge), the liquidation engine may not have time to close the position when "margin = 0" – by the time the position is closed, the price has already fallen below, and the loss exceeds margin. The role of the insurance fund: to fill the gap in positions and ensure that other users are not affected. Source: initial injection from the exchange + "remaining margin" at liquidation (the difference generated when the closing price is better than the bankruptcy price). Case: May 19, 2021 "519 Incident" On May 19, 2021, the crypto market experienced "Black Wednesday": Background: May 12 Tesla announced that it would no longer accept BTC payments May 18 China's three major financial associations banned financial institutions from participating in crypto businesses May 19 Panic selling reached its peak

Data for the day: BTC plummeted from \$43,000 to around \$30,000 (a drop of about 30%) ETH dropped 40% and the total liquidation amount across the network exceeded \$10 billion

Binance insurance fund changes: Before 519: ~100 million USD On 519 day: ~20 million added (liquidation of remaining margin) After 519: ~120 million USD

Why did the insurance fund increase instead? Because during the plummeting process, when some users' positions were cashed out, the actual transaction price was higher than the "bankruptcy price" (that is, there was remaining margin after the liquidation). This remainder automatically entered the insurance fund.

Lessons for users: In extreme market conditions, 10x or even 5x leverage may cause users to liquidate their positions within a few hours. On the day of 519, BTC fell from \$43,000 to near \$30,000 within 24 hours (a drop of about 30%), and users who used 3x leverage to do long positions would also be cashed out. High leverage should be avoided in highly volatile markets, and stop-loss orders and position limits should be set in advance.

Another case: the "312 Incident" and the ADL dispute on March 12, 2020. On March 12-13, 2020, Bitcoin plummeted by about 50% in two days (from around \$8,000 to \$3,800). What is ADL (Automatic Deleveraging)? When the insurance fund is insufficient, the exchange will initiate a "last resort" – forcibly reducing the positions of high-profit users to fill the gap of loss-making users. BitMEX's ADL incident: BTC plummeted on March 12, and a large number of long positions were cashed out. However, market liquidity dried up, and liquidation orders could not be completed. BitMEX's liquidation amount increased sharply. Insurance funds were under pressure, and the platform may trigger ADL: forced reduction of highly profitable short positions.

Users affected by ADL were angry: "I was profitable, so why was my position force-closed?" "The ADL execution price was unfair and far below the market price." "BitMEX is manipulating the market." The controversy is that ADL protects the exchange and losing-side users, but it can harm profitable users. Leading platforms usually rely first on insurance funds, liquidation optimization, and risk thresholds to reduce the probability of ADL, but automatic deleveraging cannot be ruled out entirely in extreme market conditions.

User Tip: Most exchanges display an ADL risk indicator (usually a bar from 1 to 5) on the user interface. The indicator will rise when your position is more profitable and the counterparty is severely shorted. When you see the indicator rising, you can proactively reduce your position to avoid being forced ADL.

7.2 Common Market Manipulation Tactics

Market manipulation is especially common in crypto because regulation is still uneven, liquidity is fragmented, and retail participation is high. The following are four common manipulation tactics. Legal, risk management, and communications teams should understand them so they can assess negative sentiment, user complaints, and potential rights-protection incidents more effectively.

① *Wash Trading - "Trade with yourself"*

What is wash trading? The same person, or related accounts, places buy and sell orders at the same time and trades with themselves. Volume appears high, but there is no genuine counterparty demand.

Why do this?

- Increase trading volume: rank higher on data websites such as CoinMarketCap
- Attract users: Exchanges with large trading volumes appear more “active” and are easier to gain trust
- Meet the conditions for listing: Some exchanges require tokens to reach a certain trading volume before they can continue to be listed.

How to identify?

- Repeated transactions from the same IP or device
- The transaction time and amount are regular (such as a transaction every 5 minutes, with the same amount each time)
- There is a serious mismatch between trading volume and the number of active users

Research and regulatory materials have repeatedly pointed out that there are significant abnormal trading volumes in some low-regulation markets; however, the specific proportion will vary greatly depending on the platform, sample and identification method, and it is not appropriate to use a single number to summarize the entire industry.

② *Pump and Dump: Pump First, Then Dump*

What is Pump and Dump? The manipulator first buys a small currency in large quantities at a low price, and then spreads good news through social media (such as Telegram groups and Twitter) to attract retail investors to follow suit. After the price is pushed up, manipulators sell at high prices to make profits, and retail investors are trapped.

Typical process: The first stage: building a position – the manipulator quietly buys a large number of tokens at a low price (usually choosing tokens with small market capitalization and poor liquidity) The second stage: building momentum – publishing "inside information" in the Telegram group, claiming that a certain token is about to skyrocket. The third stage: pumping up – retail investors follow the trend and buy, and the price rises rapidly. The fourth stage: shipping – the manipulator sells at a high level, and the price plummets. The fifth stage: returning to zero – Retail investors were trapped, and the token price fell back to its original point or even zero.

How to prevent it?

- Be wary of “pumping” tokens – if an unknown token suddenly skyrockets 10x, it’s likely a pump and dump

- Don't trust "insider information" in Telegram groups
- Pay attention to the concentration of token holdings – if the top 10 addresses hold 90% of the tokens, the risk of manipulation is extremely high

③ *Spoofing: Place Orders, Then Cancel Them*

What is spoofing? The manipulator places large buy or sell orders on the order book, creating the illusion that "many people want to buy/sell" and induce other traders to follow suit. After other traders enter the market, the manipulator quickly cancels the order and makes a profit by operating in the opposite direction.

For example: The manipulator placed a buy order for 100 BTC at \$60,000. Other traders saw "someone wants to buy 100 BTC" and thought the price was going to rise, so they followed suit and bought. The price was pushed up to \$60,500. The manipulator withdrew the buy order for 100 BTC and made a profit by selling at \$60,500.

How to identify?

- Large orders appear frequently and disappear quickly
- There is a serious mismatch between order book depth and actual trading volume
- A large number of orders placed and canceled in a short period of time

④ *Layering: Creating a False Sense of Depth*

What is a layered entry order? The "upgraded version" of spoofing – the manipulator places a large number of fake orders at multiple prices at the same time, creating the illusion of a "thick order book" and making other traders think that the market liquidity is very good.

The difference between spoofing and spoofing: spoofing is placing one or two large orders, while tiered orders are placing a large number of small orders at multiple prices at the same time.

How do exchanges respond?

- Abnormal order monitoring: detect frequent order placement and cancellation behavior
- Order book abnormal density alarm: trigger an alarm when the order volume at a certain price suddenly increases
- Punishment mechanism: restrict transactions or freeze funds on accounts confirmed to have manipulation behavior

Commonly used monitoring indicators on the exchange side include:

indicator	Purpose
self-transaction rate	Identify the same entity or related accounts to complete the transaction yourself

indicator	Purpose
Order placement and cancellation ratio	Identify the fraudulent behavior of placing high-frequency orders and quickly canceling them.
Account association map	Identify account groups with the same device, same IP, and same fund source
price deviation	Identify a trading pair whose price deviates significantly from the external market
Transaction concentration	Determine whether the transaction is controlled by a few accounts
API high frequency exception	Identify programmatic manipulation and abnormal order behavior
Project team / market-maker account isolation	Prevent conflicts of interest among project parties, market makers and internal accounts of the platform

Practitioner note: As an ordinary user, the most effective preventive measures are: (1) Do not chase the rise of unknown small coins; (2) Do not trust "insider information" on social media; (3) Use limit orders instead of market orders to avoid being manipulated to take advantage of slippage. Summary of this chapter

- Leverage amplifies returns and risks in equal proportions, and the design of the liquidation mechanism directly affects user trust.
- Although market manipulation takes many forms, it can all be summed up in the end: creating false supply and demand signals.
- The risk management system needs to continuously find a balance between "protecting users" and "not accidentally damaging normal transactions"

Chapter 8 Business Model and Financial Management

Chapter Overview

The income of exchanges is highly dependent on market conditions – in bull markets, they make a lot of money, but in bear markets, they may not be able to make ends meet. This chapter is about what an exchange’s “financial statements” look like.

8.1 Revenue Structure

Exchange revenue usually comes from trading fees, value-added services, interest, staking and custody income, data services, listing-related services, and institutional business. Platform mixes differ widely. For example, Coinbase’s 2023 trading revenue and subscription-and-services revenue were already of a similar scale, while some offshore full-service exchanges may still rely more heavily on spot and derivatives fees. Five factors shape revenue: market conditions, user scale, product breadth, fee competitiveness, and market share. A bull market can sharply increase volume and revenue; a bear market can shrink both. User scale sets the ceiling for potential volume. Derivatives often generate more revenue than spot. Low fees can attract large traders but compress margins. Market-share rankings must be read carefully because spot, derivatives, real liquidity, visits, and wash-trading filters use different methodologies. Exchange revenue is like a seaside resort: trading fees are ticket sales, while value-added services are restaurants, hotels, and entertainment.

8.2 Key Financial Indicators

In addition to trading volume and fee income, exchange operators also need to pay attention to the following core financial indicators:

LTV/CAC (user lifetime value/customer acquisition cost)

- LTV: The total fee income a user contributes to the exchange during the entire life cycle
- CAC: The average cost to acquire a user (advertising, referral commissions, registration rewards, etc.)
- Health benchmark: $LTV/CAC \geq 3$ (that is, for every \$1 invested in customer acquisition cost, the expected recovery is more than \$3)
- Industry reference: LTV/CAC of leading exchanges is usually between 3-5, while small and medium-sized exchanges may be as low as 1-2
- LTV/CAC is a core metric for measuring growth health. If $CAC = \$100$ and $LTV = \$50$, then growth means “the more you grow, the more you lose money.”

ARPU (average revenue per user)

- Average monthly ARPU = total monthly revenue / number of monthly active users (MAU)

- Industry reference: The average monthly ARPU of leading exchanges is about \$5-\$15, and that of small and medium-sized exchanges is about \$1-\$5.
- Fluctuations in ARPU can reflect changes in user quality (new users vs old users, retail investors vs institutions)

Gross profit margin (GM)/net profit margin

- Gross profit margin = (total revenue - direct costs) / total revenue
- The exchanges disclose less (most of them are unlisted), and the data of listed companies can only be used as a reference and cannot be directly applied to offshore exchanges or derivatives platforms.
- Net interest rates fluctuate greatly: the difference between bull and bear markets can reach more than 10x

Cash reserves and months of operation (Runway)

- Cash reserves should be reserved in bull markets to cope with bear markets - cash reserves need to cover 12-24 months of operating costs
- Calculation formula: Runway (month) = cash / average monthly operating expenses
- Industry Lessons: FTX went bankrupt with serious liquidity and balance sheet gaps

Key Metrics Cheat Sheet:

indicator	health benchmark	red flag
LTV/CAC	≥ 3	< 1.5
Average monthly ARPU	\$5-\$15 (head)	$< \$1$
Cash Runway	≥ 12 months	< 6 months
Volume share	Top 10	Continuous decline

8.3 Cost Structure: Where Exchanges Spend Money

Opening an exchange is not simply a matter of “setting up a system and collecting fees.” Operating costs are substantial. The following are the main cost categories, shown as typical shares of total exchange operating cost:

Technical infrastructure: 25-35% - Servers and cloud services (matching engine, market quotes, wallets, etc.) - Security protection (DDoS defense, penetration testing, security audit) - System development and maintenance (front-end, back-end, mobile terminal)

Labor cost: 30-40% - Technical team (development, operation and maintenance, security) - the largest head - Operation team (customer support, market, community) - Compliance team (legal affairs, KYC/AML review) - Management

Compliance and Legal: 5-15% - License application and maintenance fees - Legal advisory fees - Audit fees - Compliance tools (Chainalysis, Elliptic, etc.)

Marketing: 10-20% - Digital advertising (Google, Twitter, YouTube) - KOL cooperation and community operations - Event sponsorship - Recommended rebate expenses (the actual rebate ratio of some exchanges is higher, and different platforms have large differences in accounting treatment of rebates and front-end and back-end reconciliation methods. Exchange financial data are highly real-time, and the static analysis methods of the three traditional reports cannot be fully applied)

Others: 5-10% - Office space - Insurance (such as D&O insurance, cybersecurity insurance) - Reserve injection (insurance fund) Real data reference: Coinbase's full-year operating expenses in 2023 will be approximately US\$3.7 billion, of which technology and development, sales and marketing, general administrative expenses and other expenses constitute a larger component. Even in a bear market, investment in technology, security and compliance cannot be significantly reduced - system stability and asset security are the lifeblood of exchanges.

Financial Management Bottom Line: Customer assets and company assets must be kept separate. What users deposit forms is a platform liability, not cash that the exchange can use at will. One of the core problems of FTX is that the boundaries between user assets, associated market makers and the platform's own funds are broken.

8.4 Bear-Market Survival: How to Make It Through Winter

The exchange's revenue is highly dependent on market conditions. When a bear market hits, trading volume can shrink by 70%-90% and revenue plummets. How to survive is a problem that every exchange must face. Typical responses of exchanges in bear markets: Case: Divergence in the 2022-2023 bear market Coinbase: Layoffs of 18% (about 1,100 people) in 2022 Another 20% of layoffs (about 950 people) in 2023 but retaining core technology and compliance teams Result: Become profitable in Q4 of 2023, successfully survive the bear market

Crypto.com: 20%-25% layoffs in 2022, significant reduction in user rewards Result: stagnant user growth, but survival

Small and medium-sized exchanges: Many closed down or were acquired. Some exchanges stopped paying or lost contact due to broken capital chains. Result: The bear market is a period of industry reshuffle, and the weak are eliminated.

Bear market survival checklist: (1) Can cash reserves cover operations for more than 12 months; (2) Whether the core team is retained during layoffs; (3) Whether there are non-trading-dependent revenue sources (Staking, custody, etc.). For exchanges where all three indicators are negative, the survival probability of a bear market is significantly reduced.

Chapter Summary

- Exchange income is highly dependent on market conditions, and costs and profits fluctuate by more than 10x between bull and bear periods.
- Bear market viability is determined by three factors: cash reserves, core team retention, and revenue diversification
- The core question of Earn products is not how much you can earn in a bull market, but whether you can survive a bear market.

Chapter 9 Exchange Case Studies and Future Trends

Chapter Overview

This chapter sorts out the competitive landscape of global exchanges, major platform positioning and typical failure cases, and discusses several industry trends worthy of attention in the future. After reading this chapter, readers should be able to understand why the exchange industry is highly concentrated and how trust, compliance and asset security determine the platform life cycle.

9.0 Global Exchange Landscape

The crypto exchange industry has a strong concentration effect. Liquidity, brand, compliance channels, and product depth reinforce one another, so trading volume tends to concentrate on a small number of platforms. Different measurement methods can produce different conclusions, however: spot, derivatives, CEX, DEX, visits, and liquidity depth should not be treated as the same metric.

Reference: Major Exchange Landscape (Public Data as of Early 2026)

Exchange rankings change quickly with market conditions, measurement methodology, and wash-trading filters. The following table is not a real-time ranking; it is meant only to help readers understand the current market structure.

Dimensions	Representative platform	Observation points
Spot trading volume leads	Binance, MEXC, Bybit, Gate, HTX, Crypto.com, KuCoin	CoinGecko's 2026 CEX/DEX trading activity report shows that Binance is still the platform with the largest spot trading volume; MEXC, Bybit, Gate, etc. have outstanding performance in long-tail assets and high-frequency token listings
Derivatives trading volume leads	Binance, OKX, MEXC, Bybit, BitMart, Gate, Bitget	The trading volume of perpetual contracts and futures is usually higher than that of spot, and is the core source of revenue and liquidity for leading exchanges.
Compliance/Fiat Entry Representative	Coinbase, Kraken, Bitstamp, HashKey, OSL	These platforms pay more attention to licenses, banking channels, audits and institutional customers, and their product lines are usually more restrained than offshore comprehensive exchanges.
DEX/on-chain trading representative	Uniswap, PancakeSwap, Hyperliquid	DEX has entered the forefront of global trading volume in some months, especially in long-tail assets, on-chain perpetual contracts and uncustodial transactions, which have seen significant growth

When reading exchange rankings, you should look at three criteria: spot or derivatives, trading volume or depth of liquidity, and whether to filter abnormal trading volume. If you only look at a single ranking, it is easy to misjudge the platform's true strength.

Major Platform Positioning

platform	Positioning	comparative advantage	main constraints
Binance	Comprehensive global platform	Strong liquidity and product coverage	Compliance pressure is high in many jurisdictions, and customer support experience is affected by user scale
Coinbase	US Compliance and Fiat Entry Representative	Listed companies and institutions have high trust, and fiat channels are mature.	The rates are relatively high, the product line is relatively restrained, and the user experience for non-U.S. users is limited by region.
OKX	Outstanding experience in technology and derivatives	Strong derivatives, API and Web3 wallet capabilities	Brand coverage and partial market access are affected by region
Bybit	Derivatives and Professional Trading User Orientation	Strong trading experience, API and derivatives products	Spot asset coverage and regional access require continued observation
Bitget	Documentation and futures trading show clear patterns	The follow-up function and new user guidance are outstanding	Product line depth and institutional services still need to be continuously verified

Tip: Don't just look at rankings when choosing an exchange. Leading exchanges usually have better liquidity and more complete products, but have different rates, customer support, and compliance restrictions; small and medium-sized exchanges may list coins faster and have more activities, but have higher liquidity and asset security risks. The key is to match trading habits, asset size and risk tolerance.

9.1 The Rise of Binance: From \$15 Million to the World's Largest Exchange

The story of Binance is one of the most classic entrepreneurial cases in the crypto industry. Key timeline: July 2017 ICO raised approximately \$15 million and issued BNB tokens

In September 2017, China issued an ICO ban and exchange ban. Binance completed its "going overseas" in about two weeks: the servers were moved overseas and the team was dispersed to Japan, South Korea, Europe and other places. This speed of execution helps Binance gain an early-mover advantage compared to the longer wait-and-see and migration cycles of some platforms.

In December 2017, BTC reached an all-time high of \$20,000. Binance's daily trading volume exceeded \$10 billion and the number of users exceeded 3 million.

2018: Expansion against the trend in the bear market: - Launch of Binance Chain (own public chain) - Establishment of Binance Labs (investment department) - BNB upgraded from "fee discount token" to "ecological token"

BNB's "quarterly destruction" mechanism in 2019: Use profits to repurchase and destroy BNB every quarter → reduce supply → increase currency price, forming a flywheel of "profit → destruction → increase currency price → attract more users"

Launchpad's first IEO project in 2019: BitTorrent Token (BTT) BTT completed \$7.2 million in financing within 15 minutes. The price soared 10x after going online. Binance Launchpad formed a strong initial launch effect and market attention.

In 2020, it acquired CoinMarketCap (the world's largest market website). CMC has more than 200 million monthly visits and becomes the traffic entrance of Binance.

2020-2021 DeFi Bull Market: Launch of Binance Smart Chain (BSC) BSC attracts a large number of DeFi projects with low gas fees BNB price rises from \$30 to \$690 (all-time high)

2023 Compliance Transformation: Reaching a \$4.3 billion settlement with the US DOJ CZ resigned as CEO, Richard Teng took over Binance's four major factors for success: Timing: The bull market started in 2017, seizing the industry's explosive period Speed: Completed overseas expansion 2 weeks after China's ban, obvious first-mover advantage Token economy: BNB's burning mechanism forms a positive flywheel Execution: CZ's technical background + ultimate efficiency culture

9.2 The Collapse of FTX: A 72-Hour Crisis

The collapse of FTX is the most dramatic event in the crypto industry. Rising stage: FTX was founded by SBF (Sam Bankman-Fried) in 2019. SBF was a trader at Jane Street, a Wall Street quantitative trading company. FTX is intended as an exchange "built by traders for traders"

Innovative products will appear frequently in 2020: leveraged tokens, prediction markets, MOVE contracts, stock tokens

2021 Received US\$900 million in financing and was valued at US\$18 billion. SBF's personal net worth reached US\$26 billion. Sponsored a lot of sports and politics (FTX Arena, Super Bowl commercials)

External comments "FTX is the next Binance" "SBF is the Steve Jobs of the crypto industry" 72 hours before the collapse: November 2 (Wednesday) CoinDesk released an investigation report disclosing the balance sheet of Alameda Research (SBF's trading company): - Assets of

US\$14.6 billion – Liabilities of US\$8 billion – 5.8 billion of which are FTT tokens (tokens issued by FTX itself)

Problem: Most of Alameda's "assets" are its own tokens. If FTT prices plummet, Alameda will be insolvent. This means there may be a commingling of funds between FTX and Alameda.

On November 6 (Sunday) CZ (Binance CEO) tweeted: "Due to recent revelations, Binance will liquidate all FTT held" (Binance holds approximately \$580 million in FTT)

This Tweet Stirs Market Panic, FTT Price Plunges From \$22

Alameda CEO tweeted: "Willing to buy all FTT at \$22" trying to stabilize market confidence – but the market didn't buy it.

FTX suspends withdrawals on Tuesday, November 8th – users find themselves unable to withdraw funds. Panic intensifies further. FTT price plummets to \$5.

CZ tweeted: Binance has signed a letter of intent to acquire FTX. The market took a brief breather – but soon realized: "If FTX is really healthy, why should it be acquired?"

On November 9 (Wednesday), Binance announced that it would abandon the acquisition. Reason: "The problem is beyond our control."

FTT price plummets to \$2

SBF urgently seeks other buyers – no one takes over

On-chain data shows that there are only about US\$1 billion in assets left in the FTX wallet, while user liabilities are about US\$8.7 billion, leaving a gap of nearly US\$8 billion.

November 11 (Friday) FTX, FTX US, and Alameda Research simultaneously filed for bankruptcy protection and SBF resigned as CEO.

Successor John Ray III (Enron bankruptcy case handler) later commented: "In my career, I have never seen such a complete failure of corporate governance and such untrustworthy financial information." The subsequent investigation found: Fund misappropriation: FTX secretly transferred user funds (approximately \$8.7 billion) to Alameda; Alameda used the money to conduct high-risk transactions, and after losses, it used FTT tokens to fill its books.

Corporate governance issues include:

- No real board of directors (just SBF and two others)
- No internal audit department
- Use Slack and Signal to communicate (messages are automatically destroyed)
- Manage \$40B in Assets with QuickBooks, Small Business Accounting Software
- Executive uses company funds to buy \$300 million in Bahamas property

Personal consequences: SBF was arrested in 2023 and sentenced to 25 years in prison in March 2024 on charges including fraud, money laundering, and campaign finance law violations. Four major lessons from the FTX collapse:

- User funds must be managed independently – FTX lends user money to affiliates, which is essentially "self-financing"
- Corporate governance cannot be a mere formality – no board of directors, no audit, no risk management
- Your own tokens cannot be used as "real money" – using FTT as a reserve asset is equivalent to using self-printed money as a certificate of deposit
- You cannot rely too much on one person – SBF controls everything alone, and no one can check him. It is worth reflecting that before the collapse, FTX received investment from top institutions such as Sequoia Capital, SoftBank, and Temasek, with a valuation of up to US\$32 billion. Why do these institutions fail in their due diligence? One reason is that FTX's financial data is only available to a very small number of people, and outside investors see a carefully packaged "growth story" rather than true financial health. This reminds us: In the crypto industry, no matter how well-known investors are, they cannot replace your own judgment.

Practitioner note: FTX's story tells us that in the financial industry, "looking professional" and "being truly professional" are two different things. Compliance cannot be just superficial; it must have substantive internal controls and audits.

9.3 More Exchange Failures: Lessons from the Past

FTX wasn't the first exchange to collapse, nor will it be the last. Looking back at major exchange accidents in history can help us understand the failure paths of exchanges.

Case 3: Mt. Gox – The First Major Collapse in Bitcoin

What is Mt.Gox? The Japanese exchange, founded in 2010, was once the world's largest Bitcoin exchange – handling about 70% of the world's Bitcoin trading volume at its peak. Timeline: 2010 Jed McCaleb founded Mt.Gox (the name comes from "Magic: The Gathering Online Exchange", which was originally a website for trading game cards and later transformed into Bitcoin trading)

In 2011, McCaleb sold Mt.Gox to French programmer Mark Karpeles (Karpeles was the CEO when it later went bankrupt and was also the subject of legal prosecution). Mt.Gox became the world's largest Bitcoin exchange, accounting for 70%+ of the world's daily trading volume.

BTC price rises from \$100 to \$1,000 in 2013 Mt.Gox user base surges

In February 2014, Mt.Gox suddenly suspended all withdrawals and the website was subsequently offline.

Findings: - 744,408 BTC were stolen (about US\$450 million at the time) - Around 6% of the global BTC supply at the time - CEO Mark Karpeles declared bankruptcy - Users lost almost all of their funds

In 2015, Mark Karpeles was arrested by Japanese police on charges of tampering with financial records (not theft) and was sentenced to probation.

In 2024, ten years later, Mt. Gox began repayments to users. The repayment amount was calculated based on the Japanese-yen claim value at the time of the 2014 bankruptcy, not the later U.S.-dollar market value of BTC. BTC rose from about \$500 to over \$60,000, so even users who recovered funds missed the enormous upside. Why Mt. Gox was hacked: the root cause was chaotic hot-wallet security management. Large amounts of BTC sat in hot wallets, hot and cold wallets were not properly separated, private-key management was weak, internal monitoring was lacking, and the CEO personally controlled all private keys, creating a single point of failure.

Technical cause: transaction malleability - Attackers modified transaction signatures, changing the transaction IDs - Mt. Gox mistakenly treated the transactions as failed and resent the BTC - Result: the same BTC was sent multiple times - Although this vulnerability was not unique to Mt. Gox, its systems had no effective safeguards. Lesson:

1. 70% market share does not equal security - on the contrary, being "too big" becomes the preferred target for attackers
2. The CEO cannot control all private keys alone - multi-signature and separation of duties are basic requirements
3. Disclose immediately after being stolen - Mt. Gox concealed it for a long time after being stolen, exacerbating user losses
4. The security of exchanges requires continuous auditing - it cannot be "built once and left alone forever"

Case 4: QuadrigaCX - The CEO Died Suddenly and Hundreds of Millions Disappeared

What is QuadrigaCX? Canada's largest crypto exchange collapsed in 2018. What happened: QuadrigaCX CEO Gerald Cotten "suddenly died" while traveling in India in December 2018. Cause of death: Complications of Crohn's disease

Here's the problem: Cotten is the only person who holds the private key of the exchange's cold wallet. After his death, the assets in the cold wallet cannot be withdrawn.

Users unable to withdraw money, exchange declares bankruptcy

In 2019, the investigation found: - There are about \$190 million in user assets in cold wallets - But about \$150 million of the so-called "cold wallets" are actually empty - Cotten used user funds to conduct high-risk transactions during his lifetime - Used fake accounts to trade on his own exchange (Wash Trading) - Used user money to buy real estate, planes, and

yachts - His "death" is still controversial. Some people believe that he may have faked his death.

2020 Court investigation concluded: "This is a well-planned Ponzi scheme" Three major doubts about the QuadrigaCX incident: The CEO died "accidentally" in India - the death certificate came from a small clinic, the authenticity is doubtful The cold wallet was empty - it was claimed that there was a cold wallet of \$190 million, but the actual balance was zero There are no internal checks and balances - one person controls all funds, no board of directors, no audit Lessons:

1. "Single person control" is the biggest risk - if the CEO of an exchange can use all user funds alone, it is not an exchange, but a personal wallet
2. Death is not an "exemption card" - after the sudden death of the CEO, the exchange should have an emergency plan (such as multi-signature, key escrow)
3. When choosing an exchange, you need to look at the governance structure - is there an independent board of directors? Is there any third-party audit? Are there multiple signatures?

Case 5: US\$1.5 Billion Stolen from Bybit in 2025 - The Largest Theft in Crypto History

What is Bybit? The world's fourth largest crypto exchange, headquartered in Dubai, is known for its derivatives trading. What happened: On February 21, 2025, Bybit's ETH cold wallet was attacked while performing a routine transfer. The attacker tampered with the front-end interface of the Safe multi-sign wallet. The multi-sign signer signed malicious transactions without knowing it.

Loss: approximately 401,000 ETH plus a small amount of other tokens, worth roughly US\$1.4-1.5 billion depending on ETH price and measurement basis. This surpassed the Ronin Bridge hack of about US\$625 million and became the largest single theft in crypto history by fiat value at the time of theft. If Ponzi schemes such as PlusToken are included, rankings may differ depending on whether they are measured by ETH amount or fiat value.

Attack techniques: - Conducted by the North Korean Lazarus organization (the same organization as the Ronin attack) - Intruded into the Safe multi-signature wallet infrastructure through social engineering - Tampered with the wallet front-end interface so that signers saw normal transactions - What was actually signed was a transaction that transferred control of the cold wallet to the attacker

Bybit's response: - CEO Ben Zhou publicly confirmed the incident within 1 hour - Launched emergency withdrawal channels to handle user runs - Replenished ETH reserves through loans and large deposits - Restored most withdrawal functions within 48 hours - Cooperated with the FBI and on-chain analysis companies to track funds

Industry response: - Multiple exchanges assisted in freezing the stolen funds - Binance, OKX, etc. blocked addresses associated with the attacker - However, the Lazarus organization has transferred most of the funds through mixers and cross-chain bridges

User perspective: Within 48 hours after the attack, Bybit suspended ETH withdrawals, but withdrawals of other currencies are still available. Some users panicked and ran, causing a backlog in the withdrawal queue. Bybit resumed ETH withdrawals within 48 hours through emergency loans and large deposits, avoiding a larger run.

Core lessons:

1. Multi-signature does not mean absolute security - if the signer's device or interface is compromised, multi-signature will be useless
2. Cold wallets may also be attacked - this attack is not "private key leakage", but "signature process hijacked"
3. National-level hacker organizations are the most dangerous opponents - the Lazarus organization has the support of national resources and its attack capabilities far exceed those of ordinary hackers
4. Emergency response ability determines life and death - Bybit resumes withdrawals within 48 hours, avoiding an FTX-like collapse

Practitioner note: After the Bybit incident, the industry began to re-examine the assumption of "multi-signature security" - not only to protect the private key, but also to protect whether the information seen by the signer is authentic. Supply chain security (integrity of wallet software, front-end code) has become a new security focus.

How Exchanges Fail: A Summary

Exchange failure is usually not caused by a single point, but by a combination of several types of problems:

- Assets stolen: hot wallet, cold wallet, multi-signature process or supply chain compromised
- Misappropriation of funds: client assets, company assets and related party assets are mixed together
- Loss of governance: founders or a handful of executives have all the authority, lacking a board of directors, audits, and checks and balances
- Run collapse: Users withdraw cash intensively, and platform liquidity cannot cover short-term redemptions
- Compliance closure: The platform has been serving restricted markets for a long time, and was eventually punished by regulators, prosecuted or cut off from banks.

Practitioner note: As a user, the most important principle is not to put all your assets on one exchange. Even if a platform is large, it may still encounter security, governance, liquidity, or regulatory issues.

9.4 User Asset Protection: How to Protect Your Funds

After reading so many cases of exchange collapse, you may ask: "So how do I protect my assets?"

Principle 1: Diversify Storage

Example of dispersed storage (only used to illustrate the idea and does not constitute asset allocation advice):

Exchange A (head): 30% – for daily trading

Exchange B (head): 20% – reserve to prevent problems with A

Personal hardware wallet: 40% – Long-term holding (HODL) assets – Recommended: Ledger, Trezor

DeFi wallet: 10% – Participate in DeFi protocol – Recommended: MetaMask, Rabby

Principle 2: Enable Every Security Measure

Account security does not end with setting a password. Consider enable at least the following measures:

- Enable 2FA, preferring hardware keys or authenticator apps instead of relying solely on SMS verification codes
- Set up anti-phishing codes to identify fake emails and notifications
- Turn on the withdrawal address whitelist and limit assets to only pre-audited addresses.
- Set a withdrawal cooling-off period and delay withdrawals after changing passwords, resetting 2FA or adding new addresses.
- Regularly check login devices, IPs and authorized apps
- Delete API Keys that are no longer in use and avoid opening withdrawal permissions for API Keys
- Try to transfer long-term assets to hardware wallets, multi-signature wallets or compliant custody accounts.

Principle 3: Review Your Account Regularly

Monthly checklist: – Check whether the exchange's reserve certificate is updated – Check whether there are abnormal login records – Check whether the API key needs to be rotated – Check whether there are authorizations that are no longer used – Confirm that there are no unusual news on the exchange (such as regulatory warnings, executive resignations)

What to do if you find an exception: – Change your password immediately – Revoke all API keys immediately – Withdraw assets to your personal wallet immediately – Contact the exchange customer support to confirm the situation

Principle 4: Recognize Warning Signs

Signs that something may be wrong with the exchange:

- Withdrawals are delayed or rejected (the most serious sign)
- Frequent turnover of executives (especially compliance officers and CFOs)
- Warned by regulatory agencies in multiple countries
- Sudden launch of "high-yield Earn products" products (possibly "attracting money")
- The reserve certificate has stopped updating or there is something abnormal.
- There are a lot of complaints from the community
- The founder, actual controller or key executives have lost contact for a long time

Risk warning: If the above signals are found on the exchange, the platform exposure should be reduced in a timely manner and relevant records should be kept to avoid expanding losses due to waiting.

9.5 Future Trends for Exchanges

Trend 1: DeFi and CEX Convergence

Direction: Order matching at centralized speed and settlement in a decentralized way. Case: After dYdX v4 migrated to the Cosmos application chain, on-chain matching and on-chain settlement were realized – each order was verified on the chain, without trusting a centralized engine. The price is that the performance is still lower than that of centralized exchanges, and the commercial performance after launch did not meet expectations. US user services have been closed in 2024.

The more representative one is Hyperliquid: it also runs on a dedicated application chain. With extremely low latency and good user experience, the average daily trading volume in 2024 will continue to exceed dYdX several times, becoming a phenomenal platform in the on-chain perpetual contract track. This shows that this technical path is commercially feasible, but the ultimate user experience and liquidity depth are still the core thresholds for whether this path can become mainstream.

Trend 2: AI-Driven Intelligent Trading

AI will be used more and more in exchanges, but it is more likely to first fall into “assisted decision-making” and “operational efficiency improvement” rather than completely automating transactions for users.

- AI trading assistant: supports natural language generation of order drafts, explanation of take-profit and stop-loss, and reminder of risk parameters
- AI risk management: assists in identifying abnormal logins, suspicious fund flows, wash trading and market manipulation patterns

- AI customer support: handles standard questions and answers, material verification and work order diversion, and transfers complex issues to handbook experts
- AI Investment Research: Organize announcements, on-chain data, macro events and market indicators into structured information

AI output cannot replace investment judgment, compliance review, or manual risk management. Automated orders, futures trading, and other high-risk products must retain clear authorization rules, risk thresholds, and manual review mechanisms.

Trend 3: Regulatory Technology (RegTech)

Case: EU MiCA - the EU's harmonized regulatory framework for crypto-assets. Stablecoin-related rules will apply from June 2024, and cryptoasset service provider (CASP)-related rules will apply from December 2024; some member states have a transition period. The core value of MiCA is to establish cross-EU CASP authorization, customer asset segregation, information disclosure, complaint handling and stablecoin reserve requirements, but companies still need to pay attention to the implementation rules of the competent authorities of each member state.

Trend 4: The Web3 Wallet Ecosystem

Case: Argent wallet's "social recovery" function - after losing the private key, the account can be restored through a vote of 3-5 trusted contacts. There is no need to remember a mnemonic phrase, and losing your private key is no longer a "disaster". Trend: Account abstraction (ERC-4337) allows users to register with email and sign transactions with fingerprints - only when the user experience is close to Web2 can Web3 truly reach the public.

Chapter Summary

- The competitive landscape of the exchange industry is moving from a "technological innovation cycle" to a "system restructuring cycle"
- The rise of Binance, the collapse of FTX, and the theft of Mt.Gox—three paths point to the same lesson: trust cannot be overdrawn
- Three major driving forces for the future industry: regulatory institutionalization, professionalization of asset custody, and maturity of institutional funds and on-chain financial infrastructure

Chapter 10 On-Chain Deposits, Withdrawals, and Multi-Chain Management

Chapter Overview

On-chain deposit and withdrawal are the most basic and most frequent operations of the exchange, and are also one of the links with the most complaints from users. This chapter explains how exchanges manage assets on dozens of blockchains, as well as the technical reasons behind problems such as "deposit not received".

10.1 Multi-Chain Assets: Same Token, Different Chains

What Is Multi-Chain?

The same token can be issued on multiple blockchains. Take USDT as an example:

USDT-ERC20: runs on the Ethereum chain, the address starts with "0x"
USDT-TRC20: runs on the TRON chain, the address starts with "T"
USDT-SPL: runs on the Solana chain
USDT-OMNI: runs on the Bitcoin chain (early version, has been phased out)

Key reminder: USDT on different chains cannot be directly transferred to each other. If ERC20 USDT is sent to the TRC20 address, the assets may not be retrieved. This is one of the most common mistakes users make.

Why Issue the Same Token on Multiple Chains?

Ethereum (ERC20): The most complete ecosystem, but gas fees may rise significantly during congestion
TRON (TRC20): Transfer costs are usually low and fast, often used for small frequent transfers
Solana (SPL): Fast and low fees, but there have been network outages or performance fluctuations in history
Arbitrum/Optimism (L2): Ethereum's second-layer solution, low fees but the ecosystem is still developing

Exchanges need to support multiple chains at the same time, allowing users to choose according to their needs.

10.2 Deposit Process: How Funds Move from Chain to Exchange

Deposit Address Allocation

When you click "Deposit BTC" on the exchange, the exchange will assign you an exclusive on-chain address. The generation process of this address:

1. The exchange's wallet system generates addresses based on the HD Wallet (Hierarchical Deterministic Wallet) protocol

2. Each user is assigned an independent address (or uses the same address but with Memo/Tag distinction)
3. Users transfer assets to this address

Required Deposit Confirmations

After the assets arrive at the address on the chain, the exchange will not arrive immediately – they need to wait for "block confirmation".

What is block confirmation? The generation of each new block is a confirmation of previous transactions. The higher the number of confirmations, the lower the probability that the transaction will be reversed (double spend attack).

Confirmation number requirements for different currencies:

BTC: Usually takes 2-6 confirmations (~20-60 minutes) ETH: Usually takes 12-30 confirmations (~3-5 minutes) TRON: Usually takes 19 confirmations (~1 minute) Solana: Usually takes 1 confirmation (~400ms, but exchanges may require more)

Why do exchanges have higher confirmation requirements than on-chain ones? Because the exchange handles larger amounts of funds, it requires a higher safety margin. For example, 6 confirmations are generally considered safe enough on the BTC chain, but large deposits may require more confirmations.

Deposit Monitoring System

The exchange's deposit monitoring process:

1. Blockchain nodes continuously scan for new blocks
2. Detect whether there is a transaction transferred to the exchange address
3. Wait for the required number of confirmations to be reached
4. Add balance to user account
5. Update deposit records visible to users

Frequently Asked Questions: Several reasons why the deposit does not arrive.

- Insufficient number of confirmations: Still waiting for block confirmation, just wait patiently
- Wrong chain selection: TRC20 was used to send to the ERC20 address (the asset may be lost)
- Memo/Tag omission: Some currencies (such as XRP, EOS) need to fill in the Memo. Omission will result in the deposit not being automatically matched to the account.
- On-chain congestion: Under extreme market conditions, on-chain trading volume surges and confirmation times are significantly extended.
- Exchange maintenance: The exchange has suspended the deposit service of this currency.

10.3 Withdrawal Process: How Funds Move from Exchange to Chain

Withdrawal Review Mechanism

Withdrawals are assets transferred from the exchange, and the review is much stricter than deposits:

1. User submits withdrawal application (address, amount)
2. Automatic check of risk management system:
 - Whether the address hits a blacklist, sanctions list, or high-risk on-chain label
 - Whether travel rule, proof of address or beneficiary information requirements are met
 - Whether the amount exceeds the limit (single transaction/daily/monthly)
 - Whether anti-money laundering rules are triggered (large transfer in a short period of time, abnormal source of funds, association with fraudulent addresses)
 - Whether the account security status is normal (2FA, login IP, device, recent password or 2FA changes)
3. Small withdrawals: automatically approved and directly broadcast on the chain
4. Large withdrawals: require handbook approval (multiple people with multiple signatures)
5. On-chain broadcast: sending transactions to the blockchain network
6. Waiting for block confirmation
7. Withdrawal completed

Hot-Wallet Funding for Withdrawals

Where do the funds that users withdraw come from?

Prioritize the use of the hot wallet balance. When the hot wallet is insufficient, allocate from the warm wallet or cold wallet. Large withdrawals may require handbook transfer from the cold wallet (involving multi-signature approval, which takes a long time)

Withdrawal Fees

There are two modes for exchanges to charge withdrawal fees:

Fixed rate: The platform sets relatively fixed network fees for some main chains, and adjusts them regularly based on on-chain costs. Floating rate: Dynamically adjusts based on the current gas fee on the chain, and adds service costs if necessary.

Practitioner note: Choosing the right chain when withdrawing money can save trading fees, but you can't just look at cheapness. Operators should remind users to confirm whether the payment platform supports the chain, whether Memo/Tag is required, and whether there is a cross-chain bridge risk. The specific trading fee is subject to the real-time display on the withdrawal page and the platform announcement.

10.4 On-chain security management

Address Blacklists and Whitelists

Whitelist: Users can pre-set "trusted addresses". Withdrawals can only be sent to whitelist addresses. Blacklist: A list of known malicious addresses maintained by the exchange (from on-chain analysis companies such as Chainalysis and Elliptic). Withdrawals to blacklist addresses will be blocked.

On-Chain Monitoring

Exchanges need to monitor on-chain dynamics in real time:

Hot-wallet thresholds: trigger alerts when a hot-wallet balance drops abnormally, a single address exceeds a balance limit, or sweeping repeatedly fails. Large-transfer alerts: monitor unusual flows involving project teams, hackers, sanctioned entities, exchange hot wallets, and whale addresses. Deposit anomaly detection: identify dusting attacks, address poisoning, suspicious small-value contamination, and Memo/Tag anomalies. Chain-fork monitoring: suspend deposits and withdrawals during forks to prevent double-spend attacks. Address database maintenance: continuously update blacklists, whitelists, sanctioned addresses, ransomware addresses, and on-chain risk labels.

Chain Fork Handling

What is a chain fork? The "split" of the blockchain – one chain splits into two chains.

The exchange's processing flow in the event of a fork:

1. Before the fork: Deposits and withdrawals of this currency are suspended.
2. During the fork: Monitor the status of both chains
3. After the fork: Decide whether to support new chain tokens
4. If supported: issue new chain tokens to users according to snapshot ratio
5. Resume deposits and withdrawals

Case: The fork of Ethereum from PoW to PoS ("The Merge") in 2022 produced ETHW (PoW chain) and ETH (PoS chain). Most exchanges support ETH (PoS), and some exchanges support ETHW.

Whether to issue ETHW, issuance ratio, snapshot time and deposit and withdrawal arrangements are handled by each platform according to announcements. Not all platforms support new chain assets.

10.5 Daily Challenges in Multi-Chain Operations

Gas Fee Management

Exchanges need to process thousands of transactions on dozens of chains every day, and gas fees are a huge expense:

Need to estimate the gas fee trend of each chain

Process withdrawals in batches during low gas fee periods

Reserve enough native tokens as Gas for the hot wallet (for example, ETH is required as Gas on the ETH chain)

Node Operations and Maintenance

Exchanges usually need to configure nodes, indexing services and monitoring services for each supported chain:

Self-built nodes: highly controllable, suitable for core assets and large-scale businesses, but high costs for hardware, synchronization, upgrades and security maintenance. Third-party

node services: fast online, controllable costs, suitable for long-tail assets or backup channels, but there are risks of service provider failure, current limiting and data

consistency. Active-standby switching: the core chain is configured with at least multiple nodes and multi-regional backups to avoid deposit delays caused by single node lagging.

Upgrade management: chain hard forks, client upgrades, and RPC interface changes may affect deposits and withdrawals, and need to be tested and announced in advance.

Wrong-Chain Deposit Recovery Process

Incorrect deposits include selecting the wrong link, missing the Memo/Tag, depositing unlisted tokens, depositing into contract addresses, or transferring from high-risk addresses.

Exchanges should establish clear processing boundaries:

1. Collect information: TxHash, deposit chain, currency, amount, deposit address, Memo/Tag, user UID and deposit time
2. Determine feasibility: Whether the assets enter the exchange's controllable address, whether private key export, multi-signature approval or contract interaction are required
3. Risk audit: Confirm the source of funds, user identity and whether sanctions, hackers or fraudulent addresses are involved
4. Technical processing: If it can be retrieved, go through multi-signature approval and small-amount testing; if it cannot be retrieved, the reason must be clearly stated.
5. Charges and timeliness: If handbook and technical processing is required, the service fee, estimated timeliness, and non-promise results should be disclosed in the rules in advance

Practitioner note: If your deposit has not been received for a long time after recharging, the first step is to go to a block browser (such as Etherscan, Tronscan) to check whether

your transaction has been confirmed. If it has been confirmed on the chain but has not been received by the exchange, providing the transaction hash (TxHash) when contacting customer support can greatly speed up the processing.

Chapter Summary

- The deposit and withdrawal experience directly affects users' security perception and trust in the exchange.
- Multi-chain management is one of the core operation and maintenance challenges of exchanges today.
- Confirmation strategy, gas fee management, address whitelist – these seemingly technical details directly affect the security of user funds.

Chapter 11 The End-to-End Token Listing Process

Chapter Overview

Listing is one of the most important business decisions for an exchange. High-quality assets may bring about growth in trading volume and users, while problematic assets may cause user losses, reputational crises, and regulatory risks. This chapter explains in detail the complete process from project screening to post-listing monitoring.

11.1 Project Screening: Which Tokens Should Be Listed?

Token Listing Application Channels

Active application by the project team: Submit through the "Token Listing application" page on the exchange's official website. Active search by the exchange: BD (Business Development) team looks for high-quality projects in the industry. Community recommendation: Discover popular projects through community voting or recommendation mechanisms.

Due Diligence Checklist

Due diligence by exchange listing teams usually includes the following dimensions:

1. Team background – the true identity and resume of the core team members – whether there are any bad records such as fraud, cessation of operations, loss of contact or major disputes – the team's technical capabilities and industry experience – the credibility of investment institutions and consultants
2. Technical audit – whether the smart contract is open source and audited – the qualifications of the audit institution (well-known audit institutions such as CertiK, SlowMist, Trail of Bits) – whether there is a backdoor or the administrator has excessive authority – whether the token can be issued, suspended, or frozen
3. Token Economics Model (Tokenomics) – Total supply and circulation – Token allocation ratio (team, investors, community, treasury) – Unlocking (Vesting) plan – Are the tokens of the team and investors locked? – Whether there is a destruction or deflation mechanism – whether there are a large number of unlocked tokens about to be released (risk of selling pressure)
4. Community and Market – Community activity (real number of fans and interactions on Twitter, Discord, Telegram) – Whether there are real users and usage scenarios – Concentration of token holdings (proportion of holdings of the top 10 addresses) – Whether other exchanges have been launched

5. Compliance risks – whether the tokens may be recognized as securities (Howey test) – whether the project team is involved in unregistered issuance – whether there is a legal opinion – whether it involves sanctioned regions or entities

11.2 Token Listing Governance and Negotiation: Who Decides?

Listing decisions cannot be made solely by BD or the marketing team. Mature exchanges usually set up a token listing committee, whose members include at least business, compliance, legal, risk management, security, wallet, trading and customer support representatives. The committee's role is to formulate traceable review conclusions, not to endorse short-term trading volumes.

Costs and Business Arrangements Related to Token Listings

Compliance platforms should not regard "payment of token listing fees" as a core condition for listing. What is more common in practice is to form supporting arrangements around risk management, liquidity and operational resources:

Due diligence and technical costs: contract audit review, wallet access, testing, announcements, customer support training and risk management rule configuration. Margin/risk reserve: used to cover abnormal events, market manipulation, project team default or delisting processing costs, the terms need to be clear. Liquidity arrangements: the project team provides initial depth on its own or entrusts an independent market maker, and accepts market making KPI constraints. Marketing cooperation: AMA, announcements, educational content, event budgets, etc. need to comply with advertising compliance and risk disclosure requirements. Information disclosure obligations: major upgrades, contract permission changes, unlocking, hacker attacks, and team changes must be notified to the exchange in a timely manner

Negotiation points: listing timing, because project teams often want to list while market attention is high; trading pairs, such as BTC, USDT, or ETH pairs; promotional resources, such as homepage placement or push notifications; and deposit terms, including the conditions under which a security deposit may be deducted.

Conflict-of-Interest Controls

The most easily overlooked aspect during the token listing process is the management of conflicts of interest and material non-public information (MNPI):

- Employees, consultants and related parties are not allowed to use token listing information to trade in advance
- Personnel involved in token listing review should declare whether they hold the project's tokens or have a commercial relationship with the project team

- Token listing information should be managed according to the "minimum scope of knowledge" before announcement, and access records can be traced
- Isolation rules should be set for platform self-operated accounts, market making accounts, project team accounts and employee accounts.
- When signing an agreement with a project team or market maker, false transactions, inductive publicity and rebate abuse should be clearly prohibited

11.3 Technical Preparation for Token Listing

Token Integration

1. On-chain contract review: Confirm token contract address, accuracy (Decimals), standards (ERC-20/TRC-20, etc.)
2. Wallet deployment: Create a deposit address for the token in the exchange wallet system
3. Deposit and withdrawal configuration: set the number of deposit confirmations, withdrawal fees, and minimum withdrawal/deposit amount
4. Matching configuration: create trading pairs, set minimum order amount, price accuracy
5. Risk management configuration: set price limit and abnormal trading activity monitoring rules
6. Front-end display: UI adaptation for apps and web pages

Testing Process

1. Testnet verification: Complete the entire deposit, withdrawal and transaction process on the testnet
2. Internal testing: Testing by exchange insiders using small amounts of real assets
3. Grayscale is online: deposits will be opened first, then transactions will be opened, and finally withdrawals will be opened
4. Monitoring and observation: Track deposit arrival, order book depth, price fluctuations, API stability and customer support tickets
5. Rollback plan: set trigger conditions for suspension of transactions, suspension of deposits and withdrawals, cancellation of activities and announcement updates in advance
6. Fully open: Fully open after confirmation that there are no problems

11.4 Post-Listing Operations

Continuous Monitoring

Listing a currency is not “just a matter of time” - exchanges need to continuously monitor:

Abnormal trading volume: has volume surged unusually, possibly due to wash trading? Price

anomalies: are there abnormal price moves that may indicate manipulation? On-chain changes:

has the project team's wallet made a large transfer, have tokens unlocked and moved to an

exchange, or have contract permissions changed? Community sentiment: is there negative news

about the project team, such as key members leaving? Security incidents: has the project been hacked or has the contract been exploited? Market-maker performance: do spread, depth, uptime, and cancellation rate meet the agreement? Unlock calendar: are team, investor, foundation, and ecosystem-incentive token releases announced in advance?

Delisting Mechanism

Under what circumstances will tokens be delisted?

The trading volume is sluggish for a long time (such as the average daily trading volume for 30 consecutive days is less than \$10,000) The project team is found to have committed fraud or illegal activities There is a security vulnerability in the token contract The project team stops maintenance and updates Regulatory requirements

Delisting process:

1. Announcement: Publish delisting announcement 2-4 weeks in advance
2. Stop Trading: Stop trading on the announced closing date
3. Withdrawal window: Give users 1-3 months to withdraw tokens
4. Close deposits and withdrawals: After the withdrawal window is closed, the tokens are completely removed from the exchange

Practitioner note: When investing in small currencies, pay attention to which exchanges the token is listed on. If it is only listed on 1-2 small exchanges, the risk of delisting is extremely high. Tokens listed on leading exchanges usually undergo more stringent due diligence, but this does not mean they are risk-free.

Chapter Summary

- Listing coins is not about choosing “the best project”, but choosing “the asset that is most suitable for the current stage”
- Five dimensions of token listing due diligence (team, token economy, liquidity, compliance, community) are indispensable
- The delisting mechanism is equally important as the token listing mechanism – the ability to exit in an orderly manner is a sign of the maturity of the exchange

Chapter 12 Stablecoin Operations

Chapter Overview

Stablecoins are the basic settlement assets in the exchange ecosystem – almost all trading pairs are priced in stablecoins, and a large number of user deposits and withdrawals are also completed through stablecoins. Understanding the operating mechanism of stablecoins is the basis for understanding exchange operations.

12.1 What Is a Stablecoin?

Bottom line: stablecoins are crypto assets designed to maintain a 1:1 peg to another asset, usually a fiat currency such as the U.S. dollar.

Why use stablecoins?

- Most crypto exchanges do not directly support fiat trading.
- Stablecoins offer a ‘digital dollar’ for the crypto world
- Trading pairs are priced in stablecoins, making it easier for users to understand the value of assets.
- Cross-border transfers are faster and cheaper than traditional banks

12.2 Comparison of Major Stablecoins

USDT (Tether)

Issuer: Tether Limited and related operating entities. Market capitalization: USDT has long ranked first among stablecoins; the latest size should be checked against CoinGecko, CoinMarketCap, or issuer disclosures. Mechanism: fiat-reserve stablecoin. Tether states that each USDT is backed by equivalent U.S.-dollar-denominated assets. Reserve composition: cash and cash equivalents, short-term U.S. Treasuries, corporate bonds, and other investments. Supported chains: Ethereum, TRON, Solana, Avalanche, and more than 15 other chains.

Controversy:

- Tether has long focused on reserve certificates or assurance reports, and has not provided a complete financial audit of the same methodology as traditional listed companies.
- fined US\$41 million by the CFTC in 2021 (because reserve assets are not all cash)
- There has been a long-term question in the market: "Whether USDT is really 1:1 reserved?"
- Despite this, USDT is still one of the more liquid and widely used stablecoins

USDC (USD Coin)

Issuer: Circle. USDC was originally promoted by the Centre Consortium and later primarily issued by Circle. Market capitalization: it has long been among the leading stablecoins;

check current market data for the latest size. Mechanism: fiat-reserve stablecoin backed by cash and short-term U.S. Treasuries. Reserve disclosure: monthly reserve attestations by a third-party accounting firm. An attestation is not the same as a full audit of the issuer's overall financial position. Supported chains include Ethereum, Solana, Avalanche, and others.

Features:

- Compliance and transparency are high – Circle is a licensed company in the United States
- More transparent than USDT – regular public reserve certificates/assurance reports
- In March 2023, it briefly decoupled to about \$0.87 due to the Silicon Valley Bank (SVB) incident (different data sources vary slightly: some show \$0.877–\$0.885), and then recovered.

DAI

Issuer: MakerDAO (Decentralized Autonomous Organization) Market value: The scale changes with mortgage assets, stability fees and market demand, please check the latest data Mechanism: Crypto-asset over-collateralization type – users deposit ETH and other assets as collateral to cast DAI Features:

- Completely decentralized – no single issuer
- Overcollateralization – the value of collateral must exceed the amount of DAI issued (usually 150%+)
- High transparency on the chain – main collateral and debt status can be queried on the chain

Latest news: MakerDAO has completed the brand migration in 2024, officially changed its name to Sky (Sky Protocol), and launched an upgraded version of the stablecoin USDS. DAI continues to circulate, but new issuance at the protocol level is gradually migrating to USDS.

FDUSD

Issuer: First Digital Labs Market value: The scale changes with exchange support, issuance and redemption, and market demand. Please check the latest data. Mechanism: Fiat currency reserve type. Features:

- The stablecoin primarily promoted by Binance
- Eligible for zero-fee promotions in Binance trading pairs
- Relatively new, with market trust still being established

PYUSD (PayPal USD)

Issuer: PayPal (issued through Paxos Trust) Market capitalization: The scale is still developing, please check the latest market data Mechanism: Fiat currency reserve type, supported by cash and short-term US Treasury bonds Features:

- Launched in August 2023, PayPal is the first major payments company to issue a stablecoin

- Directly integrated into the PayPal and Venmo ecosystem, reaching hundreds of millions of users
- Currently mainly issued on Ethereum, the scale is still growing.
- Represents the trend of integration of traditional finance and encryption world

12.3 Stablecoin Depeg Events

USDC Depeg in March 2023

Background: On March 10, 2023, Silicon Valley Bank (SVB) collapsed. Circle holds approximately \$3.3 billion in USDC reserves in SVB.

Market reaction: On March 11, the price of USDC unanchored to about \$0.87 (different data sources show \$0.877-\$0.885). Users panic-sold USDC in exchange for USDT or other assets. The USDC/USDT trading pair on DEX experienced huge slippage.

Circle's response: Statement on March 12 confirming \$3.3 billion trapped in SVB Contacts the Fed and FDIC for solutions March 13, Fed announces full protection for SVB depositors USDC price quickly recovers to near \$1

Lessons:

- Even the most “compliant” stablecoins can become unanchored due to banking risks
- It is important for the reserve assets of stablecoins to be dispersed – they cannot all be stored in one bank
- Exchanges need to have contingency plans for stablecoin depegging

UST Collapse in May 2022

UST (TerraUSD) is an algorithmic stablecoin – it does not rely on asset reserves, but relies on algorithms and arbitrage mechanisms to maintain a 1:1 anchor.

How the crash went: May 7: UST begins to unanchor, falling to \$0.98 May 9: The unanchor accelerates, falling to \$0.60 May 10: falling to \$0.30 May 12: falling below \$0.10 Final: UST is almost at zero, with LUNA (UST's sister token) down from about \$87 before the crash Plunged to nearly zero (during this period, due to emergency issuance, the circulation expanded from about 350 million coins to trillions of coins, and the unit price lost its reference significance)

Loss: approximately US\$40-60 billion in market value evaporated, depending on the measurement basis used.

Lessons:

- The mechanism of algorithmic stablecoins may collapse under extreme market conditions

- The premise for "arbitrage to maintain anchoring" is market confidence – once confidence collapses, the arbitrage mechanism will fail
- Exchanges have since become more cautious about listing algorithmic stablecoins

12.4 Stablecoin Operations at Exchanges

The exchange's daily operations involving stablecoins include:

Deposit and withdrawal monitoring: Whether the deposits and withdrawals of stablecoins on each chain are normal
 Liquidity management: Ensure that mainstream trading pairs (such as BTC/USDT, ETH/USDT) have sufficient USDT liquidity
 Unanchoring contingency plan: When the stablecoin is unanchored beyond the threshold, relevant trading pairs are suspended or the margin rate is adjusted
 Chain selection optimization: Guide users to use chains with lower fees (such as TRC20 instead of ERC20)
 Reserve proof follow-up: Pay attention to USDT, USDC Reserve certificates, attestation reports and regulatory disclosures for other mainstream stablecoins.
 Issuer event monitoring: pay attention to bank partners, redemption channels, frozen addresses, regulatory litigation and changes in service areas.

Stablecoin risk list:

- Reserve Quality: Cash, Treasury bills, commercial paper, reverse repos, and other investments have different risks
- Issuer freezing ability: Centralized stablecoins can usually freeze specific addresses. Transactions require understanding of freezing trigger conditions and user impacts.
- Redemption conditions: Minimum redemption amount, redemption time, KYC requirements and bank working days will all affect 1:1 redemption ability
- Multi-chain risk: The same stablecoin is issued across multiple chains, and bridging, contract upgrades and on-chain congestion may cause short-term risks.
- Margin discount: When using stablecoins as contract margin or lending collateral, the mortgage rate, discount rate and anchorage disposal rules should be set
- Geographical restrictions: Issuers may restrict use by specific countries, regions or sanctioned entities, and exchanges need to synchronize access rules.

Tip: Stablecoin allocation is not a question of "which one is absolutely safe", but a trade-off between liquidity, transparency, redeemability, chain support and regulatory risk.

Operators should not make investment recommendations to users, but can remind users to avoid concentrating all stablecoin exposure on a single issuer or chain.

Chapter Summary

- Stablecoins are the infrastructure of the crypto market, and different types face risks from different dimensions – there is no "risk-free" stablecoin
- Core lessons from the UST collapse: Algorithmic stablecoins have structural vulnerabilities such as a “death spiral” under extreme market conditions

- Reserve transparency and regulatory compliance are becoming the winners and losers in the stablecoin market competition

Chapter 13 Data-Driven Operations

Chapter Overview

Chapter 5 introduced growth indicators, and this chapter goes one step further – talking about how exchanges use data to guide daily operational decisions. From user stratification to churn warning, from A/B testing to data burying, these are the "basic skills" of operators.

13.1 User Segmentation: Not All Users Are the Same

Applying the RFM Model to Exchanges

RFM is the most classic user stratification model in traditional marketing, with three dimensions:

R (Recency): How long ago was the last transaction F (Frequency): How many transactions were made in the past 30 days M (Monetary): How much trading volume/fee was contributed in the past 30 days

Based on the RFM portfolio, exchange users can be divided into the following categories:

User stratification	R (recent transaction)	F (trading frequency)	M (transaction contribution)	Operation strategy
High value active users	high	high	high	VIP services, exclusive customer support, institutional activities
High value silent users	low	low	high	Key recalls, exclusive return visits, and risk appetite reviews
Potential new users	high	high	low	Product education, rates and benefits, first transaction guidance
Users at risk of churn	low	low	low	Automated recall, troubleshooting, and experience return visits
Sleeping user	extremely low	extremely low	extremely low	Reach at low cost and avoid over-marketing

Pyramid Model

In addition to RFM, exchanges also commonly use "trading volume pyramid" layering:

Top 0.1%: Institutional clients and large investors – the proportion of trading volume may be significantly higher than the proportion of people. Upper level (1%): High net worth

individual users – contribute stable trading volume and trading fees. Middle level (10%): Active retail investors – high number of people and product feedback value. Bottom level (89%): Ordinary users – the largest number of people, which is the basis for brand reputation and long-term growth.

Key points: Exchange users tend to have an obvious "long tail + head concentration" structure. The operations team needs to pay attention to large accounts and institutions, but cannot use short-term trading volume to cover up user protection, appropriateness and product risks.

13.2 Churn Warning: Signals That a User May Leave

Churn Signal Model

Through user behavior data, it is possible to predict in advance which users are likely to churn. The following probabilities are training examples. The actual model needs to be calibrated based on platform historical data, user types and market cycles:

Strong signal (churn probability 70%+): – Not logged in for 14 consecutive days – Withdrawed all assets – Cancel all pending orders – Contact customer support to request account cancellation

Medium signal (churn probability 40-70%): – Transaction frequency drops by more than 50% – Downgrade from futures trading to spot trading – Login frequency changes from daily to weekly – Start to reduce positions (gradually clear positions)

Weak signals (churn probability 20-40%): – Browsed competitors' ads – Followed competitors' Twitter – Posted negative comments in the community – Trade amount became smaller

Win-Back Strategy

Different recall methods are adopted for users at different stages of churn:

Prevention stage (medium signal): – Push personalized content (market analysis, new feature introduction) – Issue trading fee discount coupons – Invite to participate in community activities

Recovery stage (strong signal): – Dedicated customer support phone return visit – Providing reasonable rate benefits or service compensation – Pushing product information that matches the risk level – Return visit from senior executives or institutional account managers (for top customers)

Abandonment stage (30+ days of churn): – Automated email sequences (low cost) – Low-frequency product update notifications – No more high-cost resources invested

13.3 A/B Testing - “Replacing Intuition with Experimentation”

What Is A/B Testing?

Randomly divide users into two groups: Group A sees the original version, and Group B sees the modified version. Compare the data performance of the two groups and use the data to decide which version is better.

Common A/B Testing Scenarios in Exchanges

Scenario 1: Registration process optimization Group A: Traditional 3-step registration (email → password → verification code) Group B: Simplified 1-step registration (mobile phone number + verification code) Metrics: Registration completion rate, KYC completion rate

Scenario 2: KYC process optimization Group A: Upload documents + face recognition in two steps Group B: Combined into one process (AI real-time recognition) Metrics: KYC completion rate, average KYC time

Scenario 3: First deposit guidance Group A: Show “Deposit Tutorial” after registration Group B: Show directly the “One-Click Buy Coin” entrance after registration Metrics: first deposit rate, first transaction rate

Scenario 4: Transaction interface Group A: The professional version of the transaction interface is displayed by default Group B: The simple version is displayed by default, and the professional version can be switched Metrics: first transaction completion rate, user retention rate

A/B Testing Considerations

The sample size must be sufficient: each group needs at least several thousand users to be statistically significant. Change only one variable: only change one element in each test, otherwise it cannot be attributed. The test time must be sufficient: run for at least 1-2 weeks to eliminate time deviation. Pay attention to long-term indicators: the registration conversion rate has increased, but the 30-day retention rate has decreased, then this change may not be good.

13.4 Event Tracking and Analytics Systems

Key Event Tracking

Core user behaviors that exchanges need to track:

Registration related: Registration page visit, registration start, registration completed, reasons for registration failure KYC related: KYC page visit, document upload, face recognition, KYC pass/rejection Deposit related: Deposit page visit, deposit address

generation, deposit credit, deposit failure Transaction related: transaction page visit, order start, order successful, order failed, order cancellation Withdrawal related: withdrawal application, withdrawal review, withdrawal received, withdrawal failure

Funnel Analysis

Connect user behaviors into a "funnel" and find the links with the most losses:

Registration funnel: Visit → Registration starts → Registration complete → KYC completed → First deposit → First transaction Transaction funnel: Visit the transaction page → Select trading pair → Place order → Transaction → Repurchase Withdrawal funnel: Initiate withdrawal → Pass risk management → Broadcast on the chain → Deposit

Find the link in the funnel with the highest churn rate and focus on optimization – this is the core method of data-driven operations.

Common Analytics Tools

Google Analytics / Mixpanel: User behavior analysis Amplitude: Product analysis, funnel analysis Looker / Tableau: Data visualization and BI reporting Self-built data platform: Large exchanges usually build their own data warehouse (based on ClickHouse or Snowflake)

Data Compliance and Privacy Boundaries

Exchange data operations cannot only pursue conversion rates, but must also comply with data protection and financial compliance requirements:

- Express consent: When collecting device information, behavioral data, marketing preferences and risk management data, the purpose should be stated in the privacy policy and authorization process
- Data minimization: Only collect data necessary for business, compliance and security to avoid "collecting what you can"
- Cross-border transfer: When user data, KYC documents and risk management labels are transferred across borders, they need to comply with GDPR, local privacy laws and regulatory requirements.
- Data retention: KYC, transaction records, customer support tickets and marketing data should be set up with different retention periods, and will be deleted or anonymized according to rules upon expiration.
- Model interpretability: Churn prediction, risk management interception and customer stratification cannot be completely black-boxed, and key decisions need to be reviewable and appealable
- Permission isolation: Operations, customer support, risk management and technical teams can only access the range of data required to complete their work

Practitioner note: When making operational decisions, first ask “does the data support this judgment?” “I think users will like this feature” is not as convincing as “A/B testing showed that Group B’ s first transaction rate was 15% higher.” Intuition can suggest hypotheses, but only data can test them.

Chapter Summary

- The core of data-driven operations is not "looking at data", but "using data to make decisions"
- The RFM model and user behavior funnel are the two basic analysis frameworks for exchange operations.
- The value of A/B testing is not to verify your intuition, but to disprove it

Chapter 14 Advanced Market-Making Operations

Chapter Overview

Chapter 6 introduces the basic concepts of market makers. This chapter goes into depth about the operating mechanism, risk management, and cooperation models between exchanges and market makers. This is advanced content, suitable for practitioners with a certain foundation.

14.1 Market-Making Strategies Explained

Inventory Management

The core challenge for market makers is “inventory management” - holding both BTC and USDT simultaneously and needing to maintain a balance between the two.

Example of the problem: The market maker placed buy and sell orders. When BTC plummeted, a large number of users sold → the market maker passively bought a large amount of BTC and BTC continued to fall → the BTC held by the market maker depreciated significantly.

Solution: delta-neutral hedging. While making markets, the market maker opens an equivalent short hedge on another venue, such as a futures market. Inventory limits: set upper and lower BTC inventory limits. When inventory exceeds the upper limit, lower the bid price or reduce bid size. Dynamic spread adjustment: when inventory is high, widen the ask spread, narrow the bid spread, and guide inventory back toward the target range.

Liquidity Provision Strategy

The market maker's strategy for placing orders on the order book:

Symmetrical pending orders: The quantity and spread of buy and sell orders are symmetrical - suitable for stable market conditions

Asymmetrical pending orders: Adjust the buying and selling ratio according to the market direction - suitable when there is a clear trend

Dynamic pending orders: Adjust the distance and quantity of pending orders in real time according to market volatility

Cross-Exchange Arbitrage

Market makers often place markets on multiple exchanges simultaneously:

Found that the price of BTC on exchange A is \$60,000 and that on exchange B is \$60,200. Buy on exchange A and sell on exchange B. Earn a price difference of \$200 (after deducting trading fees and gas fees)

This arbitrage also helps prices converge across exchanges.

14.2 Market-Maker Risk Management

Risk Types

Market risk: price fluctuations of held assets Inventory risk: one-way positions caused by imbalance in buying and selling Adverse selection risk: informed traders (such as insiders) use information advantages to trade with market makers Technical risk: algorithm bugs, network delays, system failures Counterparty risk: problems may occur on the exchange itself (such as the FTX incident)

Risk Management Mechanisms

Position limit: The maximum position of a single currency Daily loss limit: Suspend market making when the day's loss reaches the threshold Volatility threshold: Expand the spread or suspend quoting when the volatility exceeds the threshold Exchange risk assessment: Conduct regular risk assessments on cooperative exchanges to diversify exchange exposures

14.3 Cooperation Between Exchanges and Market Makers

Cooperation Models

Liquidity incentive plan: The exchange provides Maker rate discounts to market makers, and in some cases uses negative rates or rebate incentives. Exclusive market maker project: The exchange invites professional market makers to provide liquidity for specific trading pairs. Market maker levels: Market makers receive different levels of rate discounts based on pending order volume and spread performance.

Market-Making KPIs

When an exchange evaluates market makers, it cannot just look at trading volume. More reasonable indicators would cover depth, spreads, stability and market quality:

indicator	meaning	operational judgment
Spread	Buy one sell one price difference	The narrower the spread, the lower the user's transaction cost
Depth	Order depth	Pay attention to the transaction amount within the price range of 0.5%, 1%, and 2%
Uptime	Online quote time	Whether the market maker continues to quote prices during the main trading hours
Fill Ratio	Transaction rate	Too low may indicate that the quotes are too far from the market or that orders are frequently withdrawn.
Cancel Ratio	Cancellation rate	If it is too high, there may be issues with wash trading, occupying space, or misleading quotes.

indicator	meaning	operational judgment
Inventory	inventory exposure	Long-term unilateral inventory may affect the stability of market making
Slippage	Slippage	Use simulated large orders to evaluate real execution quality
Market Quality	Abnormal transaction proportion	Identify self-dealing, matching, wash trading and rebate arbitrage

Conflicts of Interest

There are natural conflicts of interest between exchanges and market makers:

Exchanges hope that market makers can provide better liquidity. Market makers hope to obtain lower rates and more rebates. If the exchange makes its own market, it may compete with external market makers.

Control principles:

- Market making accounts, project team accounts, platform self-operated accounts and ordinary user accounts should be clearly labeled and have permissions segregated
- The market making agreement should prohibit self-transactions, false transactions, commission rebates, induced transactions and price manipulation.
- Negative rate or rebate plans must set upper limits to prevent subsidies from being obtained through meaningless transactions.
- Even when the project team appoints the market maker, the exchange still needs to independently monitor the market maker's quotes and trading behavior.
- If the platform has self-operated market making, it should disclose the conflict of interest management mechanism and avoid using user order information.

Special reminder: One of the core lessons from the FTX collapse is that the relationship between the exchange and the market maker (Alameda) is too close. A healthy exchange ecosystem should maintain an independent and transparent cooperative relationship between exchanges and market makers.

Chapter Summary

- Market makers are outsourcers of exchange liquidity. Their core capabilities are inventory risk management and quoting strategies.
- The relationship between exchanges and market makers is evolving from simple rate concessions to more complex peer-to-peer partnerships.
- The main risks faced by market makers—inventory risk, adverse selection risk, and systemic risk—are highly overlapping with the exchange's own risks.

Chapter 15 Customer Support and Complaint Handling

SOP

Chapter Overview

Customer support is the front-line user touch point of the exchange – when users encounter problems, the processing speed and quality directly determine whether the user stays or leaves. This chapter lists the most common types of user complaints and standardized handling procedures.

15.1 Common User Complaints and How to Handle Them

1. *Deposit Not Credited (Most Common: About 30-40%)*

The user said: "I transferred BTC to the exchange, but it has been 2 hours and it still hasn't arrived!"

Troubleshooting steps:

1. Allow users to provide necessary information: UID, currency, chain name, amount, TxHash, deposit address, Memo/Tag, deposit time and screenshots
2. Check the transaction status on the block explorer:
 - The transaction is not confirmed → Inform the user that there is congestion on the chain and wait patiently
 - The transaction has been confirmed, but the number of confirmations is insufficient → inform the user that they still need to wait for N confirmations
 - The transaction has been confirmed and the number of confirmations is sufficient → Check the exchange's internal system
3. Check whether the deposit address and Memo/Tag are correct
4. Check if the correct chain is selected (like TRC20 vs ERC20)
5. If the chain is normal but the internal system has not been recorded → Submit a technical work order for investigation

Work order fields:

Field	Purpose
UID / registered email	Confirm account owner
Currency/chain name	Avoid confusing assets with the same name or multi-chain assets
TxHash	Verify transactions in block explorer
Deposit address/Memo/Tag	Determine whether to enter the exchange's controllable address
Amount / time	Matches wallet system and risk management records

Field	Purpose
Screenshot/Video	Assist in identifying user operation paths

Common reasons:

- On-chain congestion (highest proportion) → Just wait
- Memo/Tag omission or error → requires handbook verification and processing
- Wrong chain selection → Technical team may be required to intervene to retrieve
- Exchange system delay → Technical troubleshooting

2. Withdrawal Rejected or Delayed (About 20-25%)

The user said: "My withdrawal request was rejected!"

Troubleshooting steps:

1. View rejection reasons:
 - Risk management interception (address blacklist, abnormal behavior) → Explain that further review is required, but do not disclose specific risk management rules
 - Insufficient KYC level → Guide users to complete higher level KYC
 - Exceeding the withdrawal limit → Notify the user of the day/month limit
 - System Maintenance → Inform maintenance schedule
2. If manual review is required → submit to the Compliance team
3. If it has been broadcast on the chain but not received → Provide TxHash to guide users to view it in the block browser

Customer Service Note: Withdrawal risk management involves anti-money laundering, sanctions screening and account security rules, and the complete thresholds, hit labels or bypass methods cannot be disclosed to users. You can explain the need for supplementary materials, estimated processing time and appeal channels.

3. Abnormal Trading Activity (About 15-20%)

The user said: "My order was not filled!" / "My position was cashed out but the price was wrong!"

Troubleshooting steps:

1. Get order number, trading pair, order type, order time, price, quantity, equipment, network environment and screenshots
2. Check order status:
 - The order has not been executed → Check whether the price reaches the trigger condition (limit order, stop-loss order)
 - Partial deal → Check market liquidity

- Canceled → Check the reason for cancellation (handbook cancellation, system cancellation, risk management cancellation)
- 3. Liquidation disputes → Obtain market price data during the liquidation period to check whether the liquidation price is reasonable
- 4. If it is confirmed that it is a system abnormality → Submit it to the technical team for investigation and evaluation of compensation

4. KYC Issues (About 10-15%)

User said: “My KYC review keeps failing!”

Common causes and solutions:

- ID photo is blurry → guide to re-photograph
- The certificate type is not supported → inform the supported certificate types
- Facial recognition failed → Guidance to try again in a well-lit environment
- Manually rejected (expired certificate or mismatched information) → explain the specific reason
- Queuing for review → Inform the estimated waiting time
- KYB/Institutional Clients Missing UBO, Director, Authorized Signatory or Company Registration Document → Supplementary Proof of Beneficial Ownership and Authorization
- Hit PEP, sanctions, negative news or high-risk area rules → transfer to the compliance team for review, customer support does not promise to pass on its own

5. Account Security Issues (About 5-10%)

User says: “My account was hacked!” / “I lost my 2FA!”

Emergency handling procedures:

1. Freeze account immediately (to prevent further losses)
2. Verify user identity (KYC information, registered email, recent transaction records)
3. Investigate suspicious login records (IP, device, time)
4. If it is confirmed that it has been stolen → Assist the user to change password, reset 2FA, and check withdrawal records
5. If large amounts of assets are involved → Submit to the security team and compliance team
6. If 2FA is lost → guide the user to reset 2FA through KYC information
7. Preservation of evidence: Save login logs, withdrawal records, device fingerprints, IP, email notifications and customer support communication records
8. External collaboration: When it comes to theft, fraud or judicial requests, cooperate with law enforcement agencies, on-chain analysis service providers and other platforms according to the process

15.2 Complaint Escalation Mechanism

Hierarchy	Processing scope	Respond to requests	Typical actions
L1 frontline customer support	Common inquiries, basic deposits and withdrawals, KYC material supplement	Respond quickly, using standard phrases and FAQs	Collect information, preliminary investigation, and create work orders
L2 expert customer support	Technical anomalies, transaction disputes, complex deposits and withdrawals	Review evidence and coordinate with technical or wallet teams	Adjust logs, verify orders, and promote compensation evaluation
L3 Security/Compliance	Account theft, fraudulent funds, sanctions hits, regulatory complaints	Highest priority, preserve evidence	Freezing accounts, on-chain tracking, and law enforcement collaboration
Management/Public Relations/Legal Affairs	Media exposure, legal action, major complaints from top VIPs	Unify the methodology to avoid multiple responses	Announcements, legal letters, crisis communications

15.3 Sentiment Monitoring and Crisis Communications

When users post complaints on social media (Twitter, Reddit):

Immediate response: Reply to the public post within 1 hour to indicate that you have paid attention. Private message follow-up: Guide users to provide detailed information through private messages or work orders. Avoid public arguments: Do not argue with users on social media about right and wrong. Follow up after the problem is solved: After the problem is solved, invite users to update the processing status. Negative diffusion control: If the post is forwarded in large numbers, contact the public relations team to unify the response methodology.

Practitioner note: The core principles for handling user complaints are "fast, accurate and sincere" – quick response (within 5 minutes), accurate troubleshooting (don't let users describe the problem repeatedly), and sincere apology (even if it is not the exchange's fault, we still apologize for the inconvenience to the user).

Chapter Summary

- The core principles of customer support and complaint handling: "Fast, accurate and sincere" – speed, accuracy and attitude are indispensable.
- Standardized SOPs can significantly reduce the decision-making pressure and error rate of front-line customer support
- Every user complaint is a test of user trust – handling it properly is a bonus, handling it improperly is a fatal injury

Chapter 16 Industry Extension: Prediction Markets

16.1 Prediction Markets: Trading on Event Outcomes

What is traded in the prediction market is not the price of a certain currency, but whether a certain event will happen. For example: "Whether a certain candidate wins the election", "Whether the Federal Reserve cuts interest rates at a certain meeting", "Whether a certain team wins the championship", "Whether a certain project will issue coins before the end of the year".

Product mechanism: The most common product form is a binary event contract, with the result being only "yes/no". If a contract turns out to be correct, the settlement is \$1; if it is wrong, the settlement is \$0. Assuming that the current price of the "Yes" contract is \$0.62, it can usually be initially understood that the implied probability given by the market is about 62%. But this is not an exact probability, as prices are also affected by fees, illiquidity, risk appetite, manipulation risk and settlement dispute premiums.

Prediction market prices will fluctuate based on news, polls, on-chain data, macro indicators and changes in trader positions. The core value of a mature prediction market does not lie in "guessing the correct result", but in compressing scattered information and financial pressure into an observable, tradable, and quotable price.

Predict the value of the market:

- Price discovery: Converging scattered information, opinions and financial pressure into a tradable price.
- Risk hedging: Companies or individuals can hedge certain event risks, such as policy changes, interest rate decisions, and event results.
- User growth: Popular events naturally bring spreadability and are suitable for attracting new users, promoting activity and increasing platform discussion.
- Data Products: Predicted market prices can be cited as sentiment or probability indicators by the media, research institutions, and quant teams.

Operational difficulties:

- Event definition: Contract issues, judgment criteria, time boundaries and data sources must be stated in advance.
- Settlement mechanism: The results usually rely on official data sources, oracles, arbitration committees or platform rules, and users must be convinced that the settlement will not be manipulated.
- Event template review: The platform needs to review in advance whether the topic is clear, verifiable, and involves illegal or sensitive areas.
- Dispute handling: When there is a dispute over the outcome, the appeal window, arbitration process and final ruling methodology need to be clarified.

- Liquidity: A few popular events are actively traded, while a large number of long-tail events may not be noticed.
- Manipulation risk: Small-cap events are easily affected by large investors, interested parties or insider information trading.
- Compliance boundaries: In the United States, some event contracts may be regarded as derivatives governed by the CFTC; sports, politics, and election contracts may also violate state-level gambling laws, public interest restrictions, and regulatory requirements where the platform is located.

Regulatory cases:

- Polymarket: In 2022, the CFTC settled with Polymarket for providing an unregistered event-based binary options market to U.S. users. This case illustrates that even if blockchain and smart contracts are used to carry transactions, event contracts may still be recognized as derivatives by regulatory agencies.
- Kalshi: Kalshi chose to operate event contracts in the United States under the Designated Contract Market (DCM) path, but continues to face border disputes at the CFTC and state levels surrounding events such as elections, political control, and sports. The core judgment in this field is that prediction markets are not simple “guessing” and will be regulated as derivatives, gambling or a cross between the two in many jurisdictions.

Chapter 17 Industry Extension: RWA and Real-World Asset Tokenization

17.1 RWA Tokenization

RWA (Real World Assets) refers to mapping off-chain real assets or their income rights, shares, and certificates to the chain, so that they can be held, transferred, mortgaged, or settled in the form of tokens. It is not about “moving a house onto the chain”, but about making a set of legal rights, custody arrangements and cash flow distribution mechanisms into identifiable assets on the chain.

Common RWA types:

- Tokenized Treasury Bonds/Money Market Funds: The underlying assets are usually short-term U.S. Treasury bonds, repurchase agreements, cash or money market fund shares. Representative projects include BlackRock BUIDL, Franklin Templeton BENJI, Ondo OUSG/USDY, etc.
- Private credit and loan assets: package corporate loans, accounts receivable, trade financing and other cash flows into on-chain shares
- Real estate and infrastructure income rights: share rents, REITs or project income rights
- Gold and Commodities: Use on-chain tokens to represent custodial metal or commodity certificates
- Stocks, funds and structured products: access on-chain trading as regulated securities or synthetic exposure

Basic operational links:

1. Assets enter legal vehicles: underlying assets are held by funds, SPVs, trusts or licensed issuers
2. Custody and auditing: banks, custodians, fund administrators or audit institutions confirm the existence and net value of assets
3. Token issuance: issuing tokens representing equity or shares through smart contracts
4. Investor access: Control the buying population based on securities laws, accredited investors, geographic restrictions and KYC/AML requirements
5. Income distribution: Interest, rent, coupon or fund income is reflected through reinvestment, net worth growth or on-chain dividends
6. Secondary circulation: transfer in permissioned chains, compliant ATS, exchange whitelist addresses or DeFi protocols

Why is it meaningful?

- For traditional finance: reduce issuance, liquidation, transfer and reconciliation costs, and improve asset programmability

- For the crypto market: Provide a more stable source of on-chain income and reduce the income model that relies purely on token inflation.
- For institutional users: RWA can be used as collateral, margin or cash management tool to improve capital efficiency
- For exchanges: RWA has the opportunity to bring new businesses such as compliant income products, institutional custody, OTC trading, margin collateral and stablecoin reserve management

Typical case:

- BlackRock BUIDL: BlackRock's first public chain tokenization fund launched in 2024. Securitize provides tokenization and transfer agency-related services. The bottom layer is mainly short-duration assets such as cash, U.S. Treasury bonds and repurchase agreements. The emergence of BUIDL marks the official entry of large traditional asset management institutions into the tokenized fund market.
- Ondo Finance: Packaging income-generating assets such as U.S. Treasury bonds and money market funds into on-chain products, and designing access restrictions for different regions and investor types.
- MakerDAO/Sky: The early inclusion of off-chain assets such as U.S. Treasury bonds into the stablecoin mortgage and income system shows that RWA is not just a trading variety, but can also become part of the balance sheet of the DeFi protocol.

How exchanges participate in RWA:

- Listing of RWA tokens: subject to completion of security attribution, investor suitability, geographical restrictions and information disclosure reviews
- Do custody and settlement: custody RWA tokens for institutional customers, and handle on-chain transfers, redemptions, dividends and reports
- Use as collateral: Allow qualified institutions to use money market RWAs as margin or collateral for lending
- Issuing income products: Combining compliant RWA with exchange Earn and institutional cash management products
- Provide OTC liquidity: quote prices for large subscriptions, redemptions and secondary transfers

Actual operational details:

- Whitelist transfers: Securities or fund RWAs can generally only be transferred between addresses that pass KYC/KYB
- Investor roster: the on-chain address needs to be consistent with the off-chain investor identity, share registration and transfer agent records
- NAV Calendar: NAV updates generally follow fund or custodian business days and may not necessarily change in real time 24/7
- Redemption deadline: On-chain tokens can be transferred at any time, but off-chain redemptions may have T+1, T+2 or longer periods

- Secondary market discount and premium: When the on-chain transaction price deviates from the net value, the exchange needs to prompt liquidity and redemption risks

Main risks:

- Legal rights risk: Whether holding tokens is equivalent to holding fund shares, claims or income rights depends on the issuance documents and local laws
- Securities compliance risks: Most RWAs have securities or fund properties and cannot be freely traded for all users like ordinary tokens.
- Custody and redemption risks: The value of on-chain tokens ultimately depends on off-chain assets, custodians, bank accounts and redemption mechanisms.
- Liquidity mismatch: The underlying assets may take T+1, T+2, or longer to be cashed out, but on-chain transactions are 24/7, and discounts may occur under extreme market conditions
- Oracle and net worth risk: RWA price depends on off-chain net worth, interest rates, redemption restrictions and data synchronization, and cannot be simply processed according to stablecoin logic.
- Geographical restrictions: The United States, the European Union, Hong Kong, Singapore and other places have different requirements for security tokens, fund sales and secondary transactions. Exchanges must impose user access and transfer restrictions.

Judgment for operators: The core of RWA is not just “putting real assets on the chain”, but the combined capabilities of off-chain legal structure, asset custody, investor access, information disclosure and on-chain settlement. RWA without a compliance structure is more like a packaged high-yield product; only RWA with a complete structure can become part of the exchange's institutional business and on-chain financial infrastructure.

Chapter 18 Industry Extension: U Cards and Crypto Payments

Chapter Overview

This chapter serves as an expansion of industry knowledge to help readers understand the basic links of U-cards, crypto payments, and stablecoin deposits and withdrawals. Most exchanges do not regard the card issuance business as a core trading system, but U-cards and payment products often connect exchange accounts, stablecoins, fiat channels, card issuers and merchant networks, thus affecting user deposit and withdrawal experience, fund security and compliance boundaries.

18.1 What Is a U Card?

U-cards generally refer to virtual, prepaid, debit or physical cards that are linked to a cryptoasset or stablecoin balance. What users see on the front end is "swipe card to spend" or "bind apple Pay/Google Pay", but behind the scenes, the payment is not made directly with BTC, ETH or USDT at the merchant, but through the card issuer, clearing network and fiat currency settlement system.

A U Card can be understood as a combined product of "crypto asset balance + fiat payment network." The user first converts crypto assets into stablecoins or fiat balance, and the partner card issuer maps that balance to a card account. When the user swipes the card, the merchant receives fiat currency, the card network and acquirer process the traditional payment transaction, and the exchange or partner platform handles balance deduction, FX conversion, and risk checks in the background.

Common forms include:

Virtual card: mainly used for online payment, subscription services and cross-border consumption. It is quick to open a card, but it is subject to many restrictions on region and merchant type.

Physical card: It can be used for offline POS consumption or ATM cash withdrawal, but the issuance cost is higher and the compliance requirements are heavier.

Prepaid card: Users deposit first and then consume, which is essentially closer to prepaid balance management

Debit card: directly linked to user account balance, instant debit when consumption is made, usually requiring more complete KYC and card issuance compliance arrangements

Practitioner note: U-card cannot simply be understood as a "tool for converting coins into cash". From an operational perspective, it is more like a cross-product of deposit and withdrawal, payment, identity review, transaction monitoring and customer support systems.

18.2 Fund Flow: How Money Moves

A U Card purchase usually passes through several layers. The user holds crypto assets or stablecoins. The platform converts or records them as a spendable balance. The card issuer opens a card account for the user. The card network handles transaction settlement. The merchant acquirer settles with the merchant. The final settlement is completed in fiat currency.

The simplified process can be broken down into six steps:

Users complete KYC/KYB on exchanges or cooperative platforms and activate card products

Users deposit USDT, USDC or other supported assets, and the platform converts them into available balances according to rules.

Users initiate credit card purchases at online or offline merchants

Card issuers and card networks verify the limit, region, merchant category code (MCC) and risk management rules

After the transaction is passed, the platform freezes or deducts the user's balance, and processes the exchange rate, trading fees and reconciliation in the background

After the liquidation is completed, the merchant receives the payment in local fiat currency, and the platform completes the update of user accounts.

The key point is this: Merchants often don't know that users initially hold cryptoassets. For merchants, this is an ordinary card payment; for the platform, it is a complex process involving asset exchange, balance accounting, payment authorization, liquidation reconciliation and exception handling.

18.3 Why Exchanges Build Crypto Payment Products

The exchange's participation in U-card and crypto payments is not mainly to replace the core trading business, but to extend the retention time of users' funds, improve the deposit and withdrawal experience, and convert stablecoin balances into daily payment scenarios.

For users: The stablecoin balance in the exchange account can be used for consumption, subscriptions and cross-border payments, reducing the friction of repeated exchanges and withdrawals.

For exchanges: it can increase account retention, payment fee income, fiat channel usage frequency and brand contacts.

For institutional clients: It can be used in scenarios such as business travel, supplier payment, freelance settlement, etc., but it must have clear financial and compliance records.

For ecological partners: card issuers, payment service providers, KYC/AML service providers and risk management data providers may all participate in the division of labor.

But operators must see the other side: once payment products reach real merchants and the fiat currency system, they will enter stricter anti-money laundering, sanctions screening,

consumer protection, chargeback and fund freezing rules. Exchanges cannot evaluate U-cards only from the perspective of “one more consumption scenario”.

18.4 Operating Focus: KYC, Limits, Risk Management, and Support

U card operation starts with access management. The platform needs to confirm the user's identity, place of residence, source of funds, risk level and purpose of use. In different regions, different card issuers, and different card network rules, the range of users allowed to open cards may be completely different.

Daily operational highlights include:

KYC/KYB: Confirm personal or corporate identity to prevent anonymous users from directly entering the payment network

Quota management: Set single, single day and single monthly consumption limits, and dynamically adjust them according to the user's risk level

MCC risk management: restrict high-risk merchant categories such as gambling, adult, sanctions, gray products, cash, etc.

Regional restrictions: Transaction interception based on user location, card issuance location, merchant location and sanctions list

FX and trading fees: clearly disclose fees for crypto conversion, cross-border spending, cash withdrawals, chargebacks, and related services so users do not discover the cost only after reading their bills.

Reconciliation and settlement: Check user balances, card issuer bills, card network settlement documents and platform internal accounts

Customer support processing: Handling high-frequency issues such as card opening failure, payment rejection, repeated deductions, refund delays, card freezing, account review, etc.

Practitioner note: U Card complaints are often more complex than trading complaints because they involve the exchange, card issuer, card network, acquirer, and merchant at the same time. Customer support cannot simply reply “completed on-chain” or “debited by the platform”; it must be able to trace payment authorization, clearing, refund, and chargeback status.

18.5 Main Risks: Frozen Funds, Fraudulent Funds, and Chargebacks

The biggest operational risk of a U Card is the tension between the anonymity and cross-border liquidity of crypto assets and the strict compliance requirements of traditional payment networks. Once risk controls are triggered by source of funds, transaction scenario, or user identity, the card account, platform balance, and even the cooperation channel may be frozen.

Common risks include:

Inflow of funds involving fraud: Funds from fraud, online gambling, money laundering or underground currency exchange enter the card account through stablecoin, and then flow out in the form of consumption or cash withdrawal.

Sanctions screening failure: The user, address, merchant or counterparty hits the sanctions list, causing the card issuer or bank to freeze funds

Chargeback and refund disputes: The user denies the transaction, the merchant fails to deliver the goods, or the payment is repeated. The platform needs to handle the dispute according to the card network rules.

Service provider withdrawal: Card issuers, banks or channel providers cease services due to regulatory pressure, and user cards may be suspended in batches

Non-transparent rates: When card opening fees, monthly fees, cross-border fees, currency exchange fees, and ATM withdrawal fees are superimposed, it is easy to cause user complaints.

Compliance risks in Mainland China: U-cards are not allowed to be packaged as a tool to circumvent regulatory restrictions, bypass foreign exchange management, or participate in virtual currency trading speculation.

Risk warning: Any product that promotes “anonymous card opening”, “free cash withdrawal”, “direct consumption of stablecoin” and “bypassing bank risk management” to users in restricted areas should be regarded as a high-risk signal. It must be jointly evaluated by the compliance, legal, payment and security teams before official operation.

18.6 Relationship to the Exchange Business

U-card and crypto payment are not the core matching system of the exchange, but they will affect the exchange's user asset flow, stablecoin usage scenarios and fiat channel capabilities. Mature platforms usually manage it offline in payment, deposit and withdrawal, institutional services or ecological products, rather than letting the trading department drive it alone.

Exchanges can participate from four angles:

As an entrance: Provide users with stablecoin deposit, exchange and card balance management

As a partner: co-design products with card issuers, payment providers, banks and compliance providers

As a risk management data source: use transaction records, on-chain address risk scores and account behaviors to assist payment risk management

As a customer support and brand undertaker: Users often attribute card problems to the exchange, so the platform needs to bear the responsibility of explanation and coordination

Judgment for operators: The value of U Card lies in connecting crypto assets and real consumption scenarios, but its success or failure does not depend on the front-end card design, but on compliance access, fund link transparency, service provider stability, risk management capabilities and customer support quality.

Chapter Summary

A U Card is essentially a hybrid product combining a crypto asset balance with a traditional fiat payment network. It is not simply a cash-out tool.

The fund flow involves exchanges, card issuers, card networks, acquirers, merchants and clearing banks. Problems in any link will affect the user experience.

The core capabilities of U Card operations are KYC, limit, MCC risk management, geographical restrictions, reconciliation and settlement, and complex customer support and complaint handling. Compliance risks are higher than ordinary trading products, and you must be especially wary of fraudulent funds, sanctions screening, chargeback disputes, service provider withdrawals, and mainland China's regulatory boundaries.

Chapter 19 Industry Extension: Public Chains, Mining, Mining Farms, and Mining Pools

Chapter Overview

This chapter serves as an expansion of industry knowledge to help readers understand the operating mechanisms of public chain consensus, mines, mining pools and PoW networks. Most exchanges are not directly involved in mining, but understanding these mechanisms can help explain the impact of block confirmations, on-chain congestion, miner selling pressure, mining pool concentration, and halving events on the market.

19.1 What Is Mining?

Basic Explanation

Mining is "using hash rate to keep accounts and get rewards."

In a network without a centralized bank or clearing house, participants need to use a set of public rules to decide who has the authority to record the next batch of transactions.

Bitcoin's solution is for miners to compete by calculating hashes: whoever finds a result that meets the difficulty requirements first will gain accounting rights, as well as block rewards and trading fees. This process is mining.

Proof of Work (PoW)

The consensus mechanism used by Bitcoin is called Proof of Work:

1. Miners collect transactions to be confirmed and package them into a "block"
2. Miners constantly try different random numbers (Nonce) to calculate the hash value of the block
3. Whoever finds a hash value that meets the conditions first (the first N bits are zero) will get the accounting right.
4. Miners who obtain accounting rights broadcast blocks to the entire network
5. After verification by other nodes, the block is added to the blockchain.
6. Miners receive block rewards (newly minted BTC + trading fees)

Mining Difficulty Adjustment

Bitcoin adjusts mining difficulty every 2016 blocks (approximately 2 weeks):

If the hash rate of the entire network increases (more miners join), the difficulty will be raised - ensuring that one block is generated every 10 minutes on average. If the hash rate of the entire network decreases (miners quit), the difficulty will be lowered

This ensures that Bitcoin's block production speed is always stable at about 10 minutes per block.

Block rewards and halving

Bitcoin's block reward is halved every 210,000 blocks (approximately 4 years):

2009: 50 BTC per block reward

2012: Halving to 25 BTC

2016: Halving to 12.5 BTC

2020: Halving to 6.25 BTC

2024: Halving to 3.125 BTC

It is expected that all 21 million BTC will be mined around 2140. At that time, miners' income will come entirely from trading fees.

19.2 Evolution of Mining Hardware

CPU Era (2009–2010): Mining can be done with an ordinary computer

GPU Era (2010–2012): Graphics card mining efficiency far exceeds that of CPU

FPGA era (2012–2013): Programmable hardware, further improving efficiency

ASIC era (2013 to present): Chips designed specifically for Bitcoin mining, with hash rate millions of times that of CPUs

Nowadays, Bitcoin mining has been completely dominated by ASIC mining machines, and it is basically impossible for ordinary computers to mine BTC.

19.3 Mining Pools: Hash-Rate Aggregation and Income Smoothing

Why Mining Pools Are Needed

As the hash rate of the entire network increases, the probability of a single miner independently mining a block is extremely low. Assuming that the hash rate of an ASIC mining machine only accounts for a very small proportion of the entire network, it may not be able to mine any blocks for a long time; and the hash rate of the entire network, mining machine performance, and electricity prices are all changing. The specific probability should be calculated based on real-time data such as mempool.space, BTC.com, and blockchain.com.

For individual miners, income is highly unstable: there may be no block rewards for a long time, or a block may be mined by chance. The role of the mining pool is to smooth low-probability large returns into relatively stable small returns.

How Mining Pools Work

Mining pools bring together the hash rate of a large number of miners to mine together:

1. Miners connect their mining machines to the mining pool's servers
2. The mining pool uniformly distributes mining tasks
3. Miners submit "Share" – proving that they are contributing hash rate
4. When the mining pool mines a block, rewards are distributed in proportion to the hash rate contributed by each miner.

The mining pool can be understood as a mechanism in which "multiple people participate together and distribute income according to their contribution": the single reward received by each miner becomes smaller, but the probability of obtaining stable income is significantly increased.

Income Distribution Models

PPLNS (Pay Per Last N Shares): Distributed in proportion to the last N shares. Earnings fluctuate more, but long-term average returns are higher

PPS (Pay Per Share): Each share receives a fixed income. Earnings are stable, but mining pools take on more risks, so fees are higher

FPPS (Full Pay Per Share): an upgraded version of PPS, the mining pool also allocates trading fees

Major Global Mining Pools

Mining pool rankings and hash rate ratios change rapidly and should be subject to real-time data websites. Common mining pools include Foundry USA, AntPool, ViaBTC, F2Pool, MARA Pool, Binance Pool, etc. It is not recommended to write down the market share of a certain date in the handbook. Operation personnel should intercept the data of that day and mark the source when they need to quote it.

The problem of hash rate concentration: The leading mining pool will control a large proportion of the entire network's hash rate in many periods. In theory, if a single entity controls more than 51% of the hash rate, it is possible to launch a double-spend attack or censor transactions. In reality, mining pools are composed of a large number of miners and cannot be simply equated to one owner. However, the concentration of mining pools is still a security and governance issue that the industry has long been concerned about.

19.4 Other Consensus Mechanisms

Proof of Stake (PoS)

Ethereum transitions from PoW to PoS ("The Merge") in 2022:

No mining machine is required. Instead, participants join consensus by staking tokens. Validators must stake 32 ETH. They are randomly selected to create blocks; honest validators receive rewards, while malicious validators are penalized through slashing.

Advantages: Extremely low energy consumption (99.95% energy saving than PoW) Disadvantages: May lead to token concentration – people holding more tokens get more verification rights and benefits

Delegated Proof of Stake (DPoS)

EOS, TRON, etc. adopt:

Token holders vote to elect 21-101 "super nodes". Super nodes are responsible for producing blocks and verifying transactions. Ordinary users indirectly participate in the consensus through voting.

Advantages: Fast speed (TRON block generation time is 3 seconds) Disadvantages: small number of nodes and low degree of decentralization

19.5 The Relationship Between Mining and Exchanges

The exchange is online or has launched a mining pool business: the coins dug by miners can be directly entered into the exchange account, which is convenient for selling, mortgage or settlement. Miners are important users of the exchange: miners need to sell the coins they mine and realize them, and are the natural sellers of the exchange. Halving events affect the market: every time BTC is halved, miners' income decreases, which may trigger selling pressure. Mining difficulty is a market indicator: the continued growth of hash rate indicates that the industry is booming; a decrease in hash rate may mean that miners are "shut down."

Practical resources: blockchain.com and mempool.space provide real-time data on the Bitcoin network, including network hash rate, mining difficulty, and block confirmation status.

Chapter Summary

- The essence of mining is "exchanging calculations for accounting rights." PoW and PoS are two different trust-building mechanisms.
- Mining pools solve the problem of excessive fluctuations in the income of a single miner, but bring new risks of centralization of hash rate.
- The energy consumption and hash rate concentration of mining will affect industry policies and public perception in the long term.

Appendix

A. Recommended Reading

Primer:

1. "Mastering Bitcoin" by Andreas Antonopoulos – the classic introductory book for understanding Bitcoin and blockchain technology
2. "The Bitcoin Standard" (Saifedean Ammous) – Understanding Bitcoin's value proposition from an economic perspective

Technical architecture:

3. Official technology blogs of exchanges such as Binance Engineering Blog and Coinbase Tech Blog – practical experience in understanding the technical architecture of exchanges
4. "Mastering Ethereum" (Andreas Antonopoulos) – Understanding Smart Contracts and DeFi

Compliance and Regulation:

5. FATF "Updated Guidance for a Risk-Based approach to Virtual Assets and VASPs" – global anti-money laundering standard
6. Official websites of regulatory agencies in various countries – track the latest regulatory developments

Industry reports:

7. CoinGecko Annual Report – Industry Data and Trend Analysis
8. CoinMarketCap Annual Report – Exchange Rankings and Market Shares
9. Chainalysis Crypto Crime Report – on-chain crime and anti-money laundering data

Case study:

10. John Ray III's Court Statement on FTX Bankruptcy (Public Document) – Learn the Details of FTX's Collapse
11. Mt.Gox Bankruptcy Documents – Learn about the major exchange mishaps in Bitcoin history

Tools and data:

12. CoinGecko (coingecko.com) – Coin prices and exchange data
13. CoinMarketCap (coinmarketcap.com) – Coin prices and exchange rankings
14. DefiLlama (defillama.com) – DeFi protocol data
15. Dune Analytics (dune.com) – on-chain data analysis

B. Key Reference Sources

The following sources were used to check regulatory, case, and trend content in this handbook. Because cryptoasset regulation and market data change rapidly, the original text and publication date should be double-checked when citing.

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7. Coinbase Fourth Quarter and Full Year 2023 Shareholder Letter:
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8. Securitize's product description of BlackRock BUIDL:
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11. CoinGecko/CoinMarketCap: Market data snapshot for market cap, volume and exchange rankings
12. mempool.space / BTC.com / blockchain.com: Real-time data for Bitcoin hashrate, block confirmations and mining pool shares

Disclaimer

This handbook is compiled based on public information and personal experience and is for internal training and research purposes only. The description of products, cases, regulatory and market data in this article does not constitute investment advice, legal opinions, tax opinions, accounting opinions, audit opinions, compliance conclusions or any form of solicitation, recommendation, or promised income.

Virtual asset trading carries high risks, prices may fluctuate violently, and derivatives, leverage, prediction markets, RWA, stablecoins and on-chain products may involve loss of principal, lack of liquidity, compliance restrictions, custody risks and technical risks. Before actually conducting business, any organization or individual should consult with qualified legal, compliance, tax, audit and security professionals based on the jurisdiction, service recipients, product types and licensing requirements.

Special reminder: Mainland China has clear regulatory restrictions on matters such as virtual currency trading and speculation, token issuance and financing, and overseas platforms providing services to domestic residents. The exchange products and operating mechanisms discussed in this handbook are only for knowledge learning and compliance research, and do not represent recommendations to conduct relevant business or participate in relevant transactions in restricted areas.

The regulatory lists, market share, market value, mining pool share, exchange ranking, product status and case progress listed in this handbook are all time-sensitive. Before formal citation, training release or external use, the latest information from regulatory agencies, issuers, exchange announcements and authoritative data platforms should prevail.

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